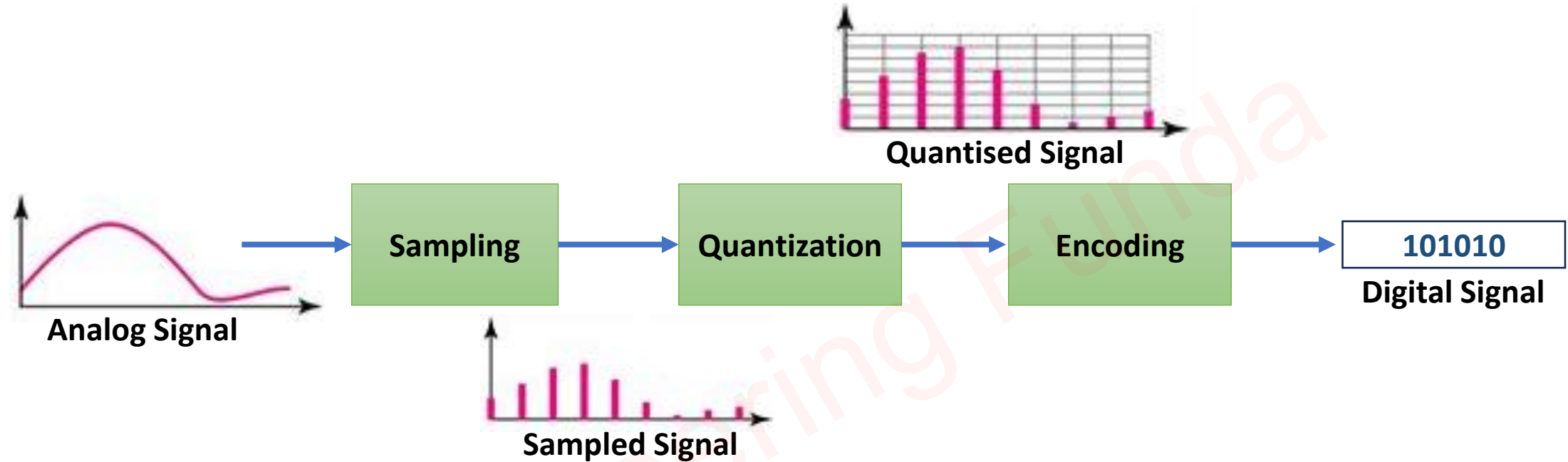


Sampling Theory

Outlines

- ☐ Analog Signal to Digital Signal Conversion
- ☐ Basics of Sampling Theory
- ☐ Waveforms of Sampling Theory
- ☐ Frequency Spectrum of Sampling
- ☐ Aliasing Effect and Nyquist Rate
- ☐ Key Steps to avoid Aliasing Effect

Analog Signal to Digital Signal Conversion



- ❑ Sampling converts a Continuous-Time Signal into a Discrete-Time Signal.
- ❑ Quantisation converts a Discrete-time Signal into a Discrete-Time Discrete-Amplitude Signal.
- ❑ Encoding gives Digital values to each amplitude level with respect to samples.

Basics of Sampling Theory

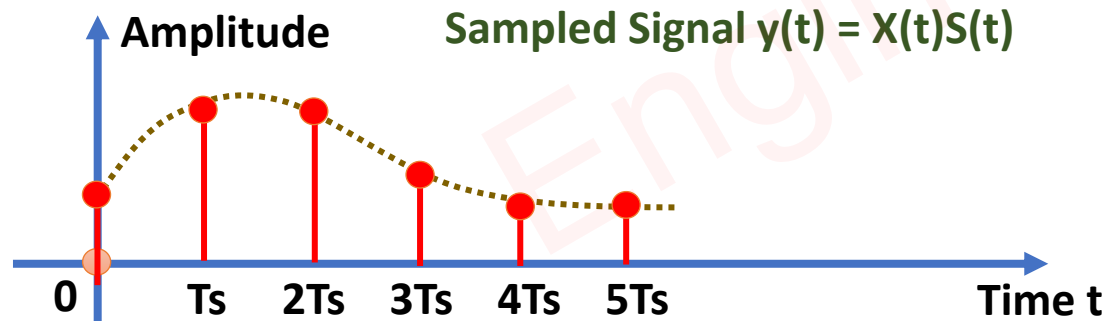
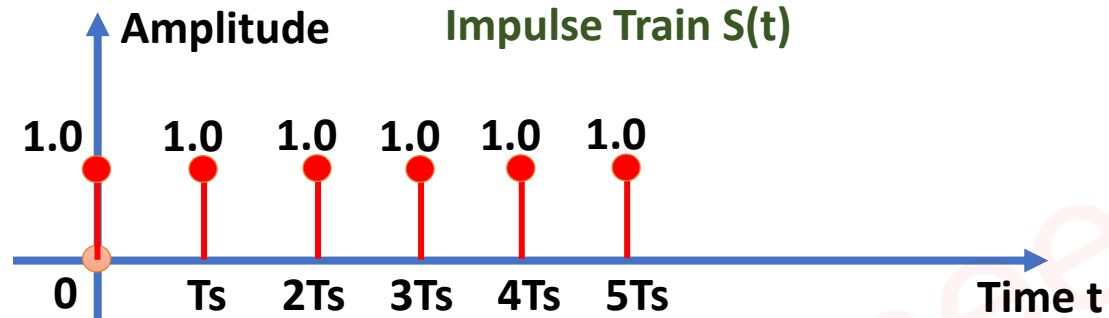
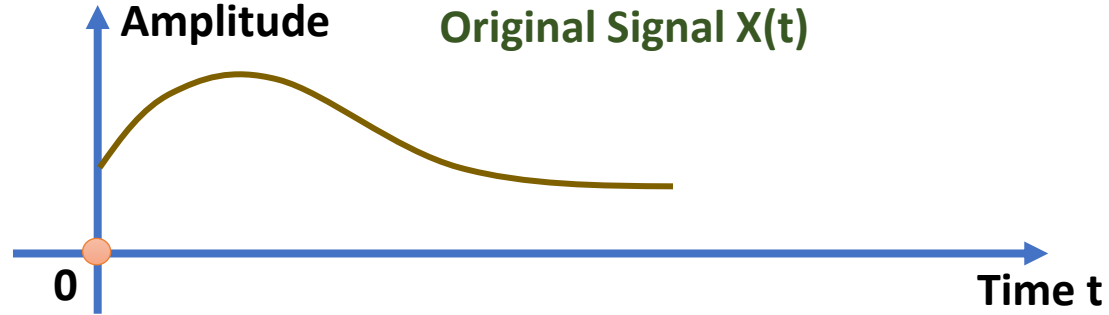
- ❑ Sampling is a Process to convert a Continuous-Time Signal into a Discrete-Time Signal.
- ❑ A sufficient number of samples must be taken to ensure that the original signal can be accurately reconstructed.
- ❑ The required sampling rate is determined by the maximum frequency present in the original signal and should be greater than the Nyquist rate.

$$\Rightarrow F_s \geq 2F_m$$

- F_s is Sampling Rate {Sampling Frequency}
- F_m is Maximum Frequency of Original Signal

- ❑ Types of Sampling
 - Ideal Sampling
 - Natural Sampling
 - Flat-Top Sampling

Waveforms of Sampling



□ Statement of Sampling Theory

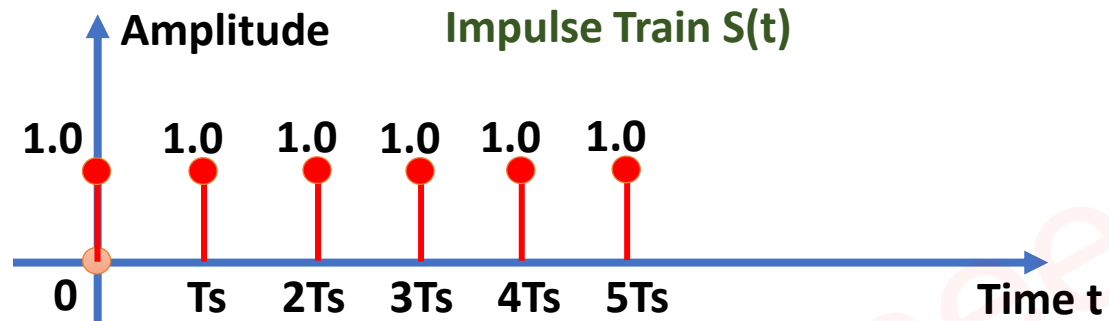
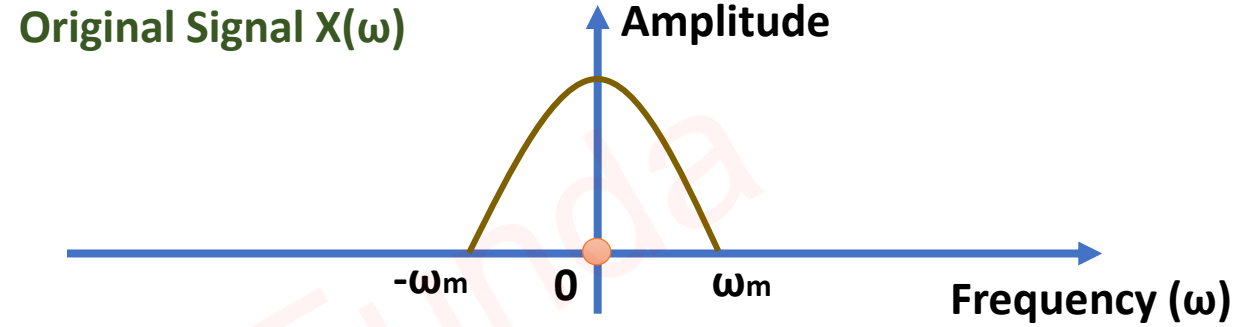
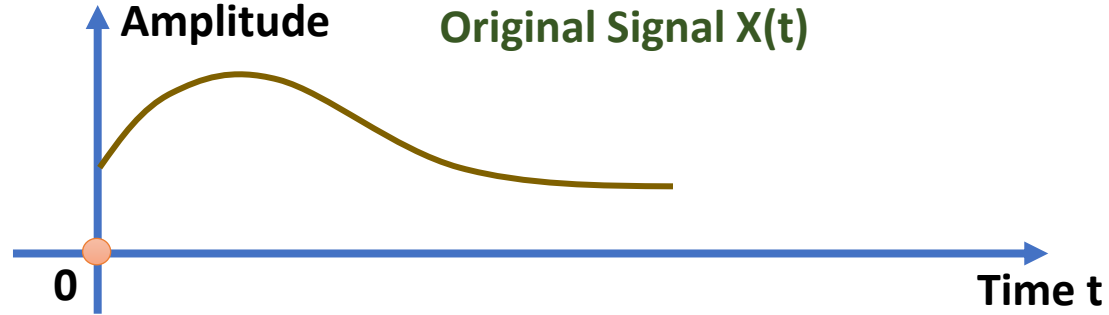
- A bandlimited signal with finite energy, having no frequency components higher than f_m Hz, can be completely described by its sample values taken at uniform intervals equal to or less than $\frac{1}{2f_m}$ seconds.

$$\Rightarrow T_s \leq 1/2F_m$$

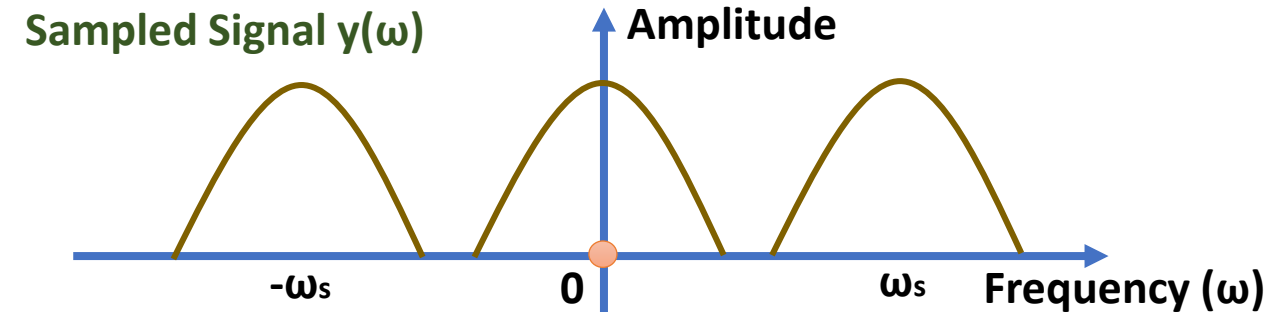
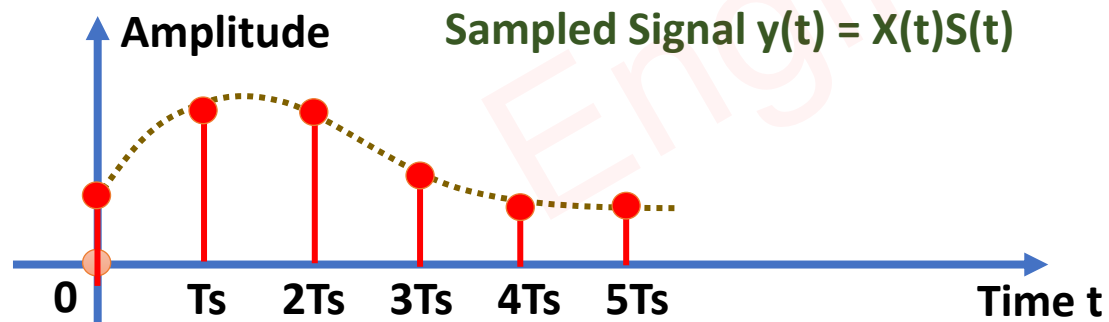
- A bandlimited signal with finite energy, containing no frequency components higher than f_m Hz, can be completely recovered from its samples if they are taken at a rate f_s that is greater than or equal to $2f_m$.

$$\Rightarrow F_s \geq 2F_m$$

Frequency Spectrum of Sampling

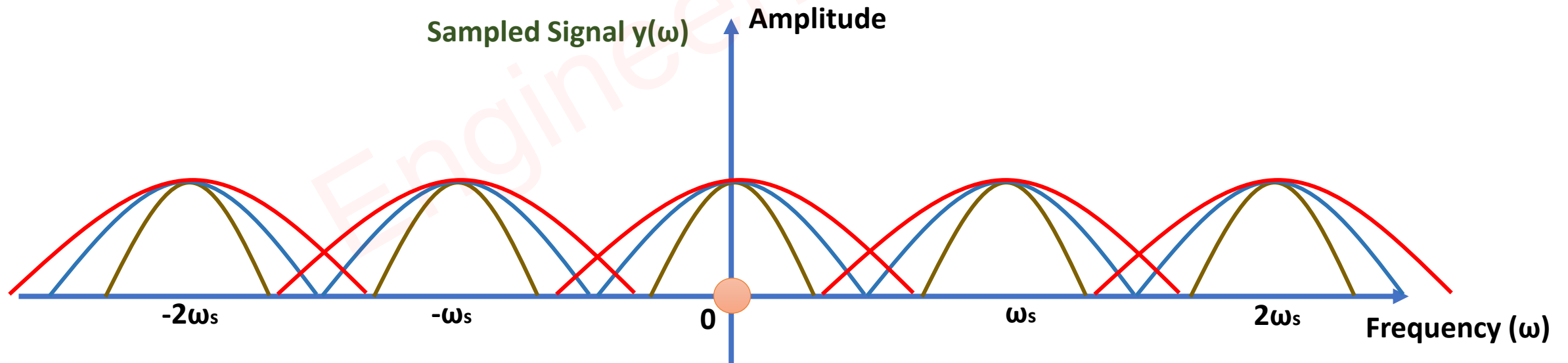


$$\Rightarrow s(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_s) = \frac{1}{T_s} [1 + 2 \cos(\omega_s t) + 2 \cos(2\omega_s t) + \dots]$$
$$\Rightarrow y(t) = \frac{1}{T_s} X(t) [1 + 2 \cos(\omega_s t) + 2 \cos(2\omega_s t) + \dots]$$
$$\Rightarrow y(\omega) = \frac{1}{T_s} [X(\omega) + X(\omega \pm \omega_s) + X(\omega \pm 2\omega_s) + \dots]$$



Aliasing Effect and Nyquist Rate

- ❑ Aliasing is a phenomenon that results from under-sampling and is also referred to as foldover error or under-sampling.
- ❑ Lets Consider Three Cases to understand Aliasing Effect:
 - Sampling Frequency is greater than Nyquist Rate ($F_s > 2F_m$)
 - Sampling Frequency is Equal to Nyquist Rate ($F_s = 2F_m$)
 - Sampling Frequency is less than Nyquist Rate ($F_s < 2F_m$)
- ❑ Prior to sampling, the original signal is filtered using a low-pass filter (LPF), commonly known as a pre-aliasing filter or band-limiting filter, to limit its frequency components.



Key Steps to avoid Aliasing Effect

- ❑ Step-1: Sampling Rate F_s Should be Greater than Nyquist Rate $2F_m$.
- ❑ Step-2: Use an Aliasing Filter before Sampling.
- ❑ Step-3: At the receiver, use a High Order (Sharp edge) Filter.
- ❑ Step-4: Keep Some Space in between successive bands of the sampled signal.

Properties of Nyquist Rate

⇒ Nyquist Rate

$$\Rightarrow F_N = 2F_m \text{ (Hz)}$$

$$\Rightarrow \omega_N = 2\omega_m \text{ (Rad/sec)}$$

⇒ Nyquist Interval

$$\Rightarrow T_N = \frac{1}{F_N}$$

⇒ Time Shifting property

$$\Rightarrow x(t) \longrightarrow NR = F_N$$

$$\Rightarrow x(t \pm t_0) \longrightarrow NR = F_N$$

⇒ Time Scaling property

$$\Rightarrow x(t) \longrightarrow NR = F_N$$

$$\Rightarrow x(at) \longrightarrow NR = aF_N$$

⇒ Power Scaling property

$$\Rightarrow x(t) \longrightarrow NR = F_N$$

$$\Rightarrow (x(t))^n \longrightarrow NR = n F_N$$

⇒ Differentiation property

$$\Rightarrow x(t) \longrightarrow NR = F_N$$

$$\Rightarrow \frac{dx(t)}{dt} \longrightarrow NR = F_N$$

⇒ Integration property

$$\Rightarrow x(t) \longrightarrow NR = F_N$$

$$\Rightarrow \int x(t) dt \longrightarrow NR = F_N$$

⇒ Multiplication property

$$\Rightarrow x_1(t) \longrightarrow NR = F_{N_1}$$

$$\Rightarrow x_2(t) \longrightarrow NR = F_{N_2}$$

$$\Rightarrow x_1(t)x_2(t) \longrightarrow NR = F_{N_1} + F_{N_2}$$

Examples of Sampling & Nyquist Rate

Example 1: Find Nyquist Rate and Nyquist Interval for given Signal:

$$\Rightarrow x(t) = 3 \cos(50\pi t) + 10 \sin(300\pi t) + \cos(100\pi t)$$

$$\leadsto f_1 = \frac{\omega_1}{2\pi} = 25 \text{ Hz}$$

$$f_2 = \frac{\omega_2}{2\pi} = 150 \text{ Hz}$$

$$f_3 = \frac{\omega_3}{2\pi} = 50 \text{ Hz}$$

$$\leadsto \text{Max. freq.} \\ f_m = 150 \text{ Hz}$$

$$\leadsto f_N = 2f_m = 300 \text{ Hz}$$

$$\leadsto T_N = \frac{1}{f_N} = 3.33 \text{ msec}$$

Example 2: Find Nyquist Rate and Nyquist Interval for given Signal:

$$\Rightarrow x(t) = \frac{1}{2\pi} [\cos(4000\pi t) \cos(1000\pi t)]$$

$$= \frac{1}{4\pi} [2 \cos(4000\pi t) \cos(1000\pi t)]$$

$$= \frac{1}{4\pi} [\cos(5000\pi t) + \cos(3000\pi t)]$$

$$\leadsto f_1 = \frac{\omega_1}{2\pi} = 2500 \text{ Hz}$$

$$f_2 = \frac{\omega_2}{2\pi} = 1500 \text{ Hz}$$

$$\left. \begin{array}{l} f_1 = 2500 \text{ Hz} \\ f_2 = 1500 \text{ Hz} \end{array} \right\} \leadsto \text{Max. freq.} \\ f_m = 2500 \text{ Hz}$$

$$\leadsto f_N = 2 f_m = 5000 \text{ Hz}$$

$$\leadsto T_N = \frac{1}{f_N} = 0.2 \text{ msec}$$

Example 3: Find Nyquist Rate and Nyquist Interval for given Signal:

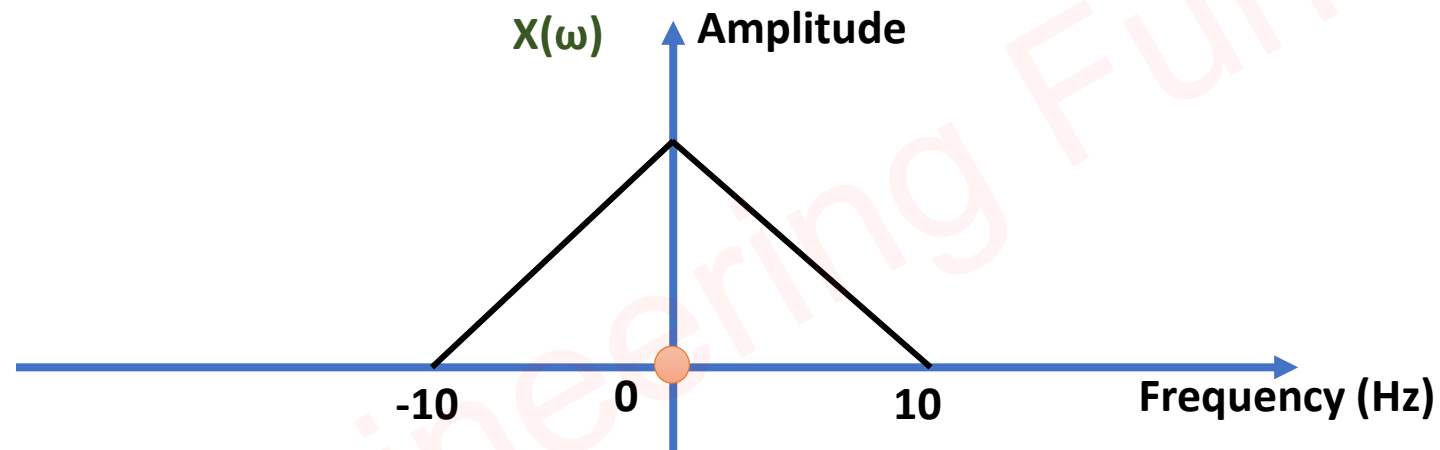
$$\Rightarrow x(t) = 6 \cos(50\pi t) + 20 \sin(300\pi t) - \sin(50\pi t)$$

$$\begin{aligned} \leadsto f_1 &= \frac{\omega_1}{2\pi} = 25 \text{ Hz} = f_3 \\ f_2 &= \frac{\omega_2}{2\pi} = 150 \text{ Hz} \end{aligned} \quad \left. \vphantom{\begin{aligned} f_1 &= \frac{\omega_1}{2\pi} \\ f_2 &= \frac{\omega_2}{2\pi} \end{aligned}} \right\} \leadsto \begin{aligned} &\text{max. freq.} \\ &f_m = 150 \text{ Hz} \end{aligned}$$

$$\leadsto f_N = 2f_m = 300 \text{ Hz}$$

$$T_N = \frac{1}{f_N} = 3.33 \text{ msec}$$

Example 4: The spectrum of a continuous-time Analog signal $x(t)$ is shown in figure. This signal is sampled at frequency F_s which is $3/2$ times the maximum frequency F_m in the spectrum of $x(t)$. Draw the spectrum of the sampled signal clearly showing the effect of under sampling. If the sampled signal is then passed through an ideal LPF having cutoff frequency $F_s = F_m$, then draw the spectrum of the signal recovered at the filter output.

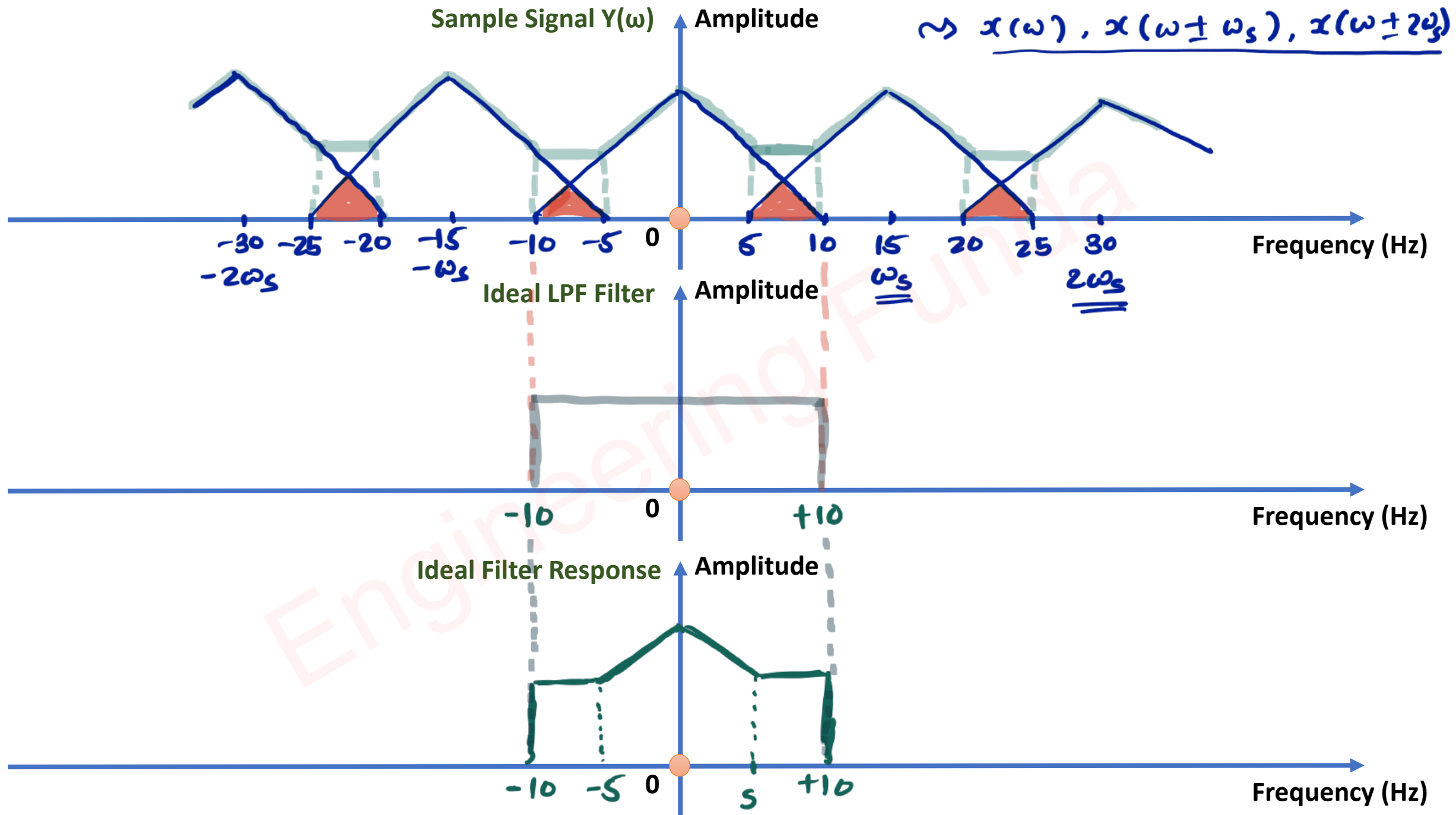


$$\leadsto F_s = \frac{3}{2} F_m = \frac{3}{2} \times 10 = 15 \text{ Hz}$$

$$\leadsto F_m = 10 \text{ Hz}$$

$$\boxed{\leadsto F_s \geq 2F_m}$$

↑
Not Followed



Examples of Sampling & Nyquist Rate

Example 1: Find Nyquist Rate {in Hz and Rad/sec} and Nyquist Interval for given Signal:

$$\Rightarrow x(t) = 2 \sin(4\pi t) \cos(2\pi t)$$

$$= \sin(6\pi t) + \sin(2\pi t)$$

$$\leadsto f_1 = \frac{\omega_1}{2\pi} = 3 \text{ Hz}$$

$$f_2 = \frac{\omega_2}{2\pi} = 1 \text{ Hz}$$

$\leadsto$ max. freq
 $f_m = 3 \text{ Hz}$

$$\leadsto F_N = 2 F_m = 6 \text{ Hz}$$

$$\omega_N = 2\pi F_N = 12\pi \text{ Rad/sec}$$

$$\leadsto T_N = \frac{1}{F_N} = 0.166 \text{ sec.}$$

$$\leadsto x(t) = 2 \sin(4\pi t) \cos(2\pi t)$$

$$\downarrow$$
$$\omega_{N1} = 8\pi$$

$$\downarrow$$
$$\omega_{N2} = 4\pi \text{ rad/sec}$$

$$\leadsto \omega_N = \omega_{N1} + \omega_{N2}$$
$$= 12\pi \text{ rad/sec}$$

Example 2: Let $x(t)$ be a signal with Nyquist rate ω_s . Determine the Nyquist Rate for each of the following signals:

i. $x(t) + x(t - 1)$

i) $\Rightarrow \underbrace{x(t)} + \underbrace{x(t-1)}$

$\downarrow$
 ω_s

$\downarrow$
 ω_s

$\rightarrow \omega_{Ni} = \omega_s$

ii. $dx(t)/dt$

ii) $\frac{dx(t)}{dt}$

$\downarrow$
 ω_s

$\rightarrow \omega_{Nii} = \omega_s$

iii. $x^3(t)$

iii) $\underbrace{x^3(t)}$

$\downarrow$
 $3(\omega_s)$

$\rightarrow \omega_{Niii} = 3\omega_s$

iv. $x(t) \cos(\omega_s t)$

iv) $\underbrace{x(t)} \underbrace{\cos(\omega_s t)}$

$\downarrow$
 ω_s

$\downarrow$
 $2\omega_s$

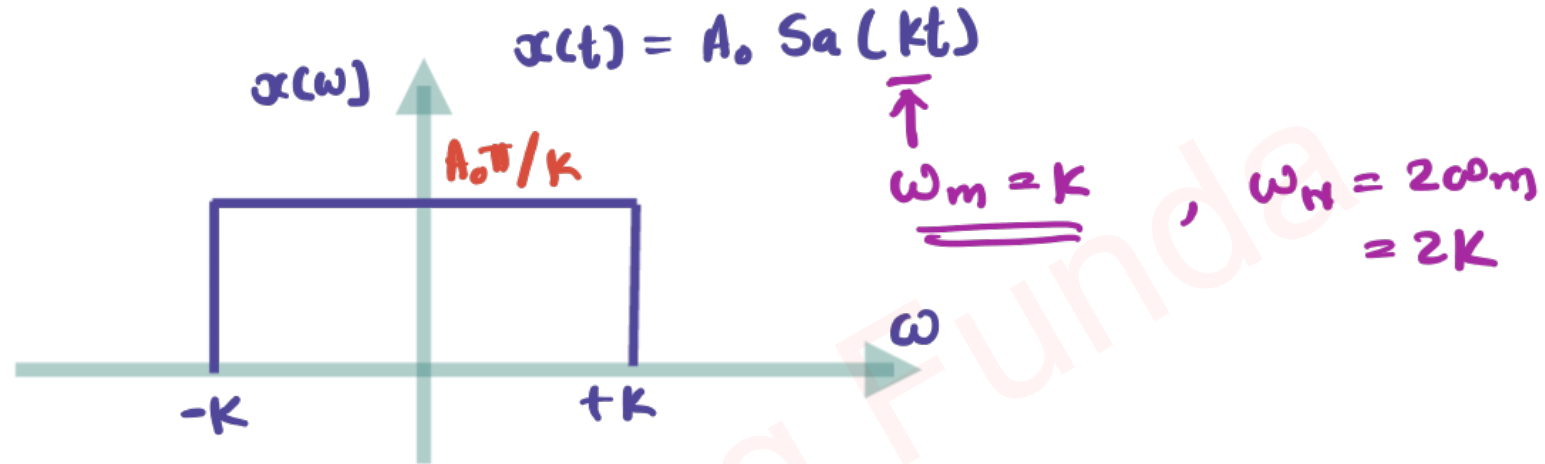
$\rightarrow \omega_{Niv} = \omega_s + 2\omega_s$
 $= 3\omega_s$

Example 3: Find Nyquist rate for each of the following signals:

i. $\text{Sa}(4\pi t)$

ii. $\text{Sa}^3(5\pi t)$

iii. $\text{Sa}^2(4\pi t)\text{Sa}^4(3\pi t)$



i) $\text{Sa}(\underline{4\pi t})$

$\downarrow$
 $\omega_m = 4\pi$

$\rightarrow \omega_{Ni} = 2\omega_m = 8\pi \text{ rad/sec}$

ii) $\text{Sa}^3(\underline{5\pi t})$

$\downarrow$
 $\omega_m = 5\pi$

$\rightarrow \omega_{Nii} = 3(2(5\pi)) = 30\pi \text{ rad/sec}$

iii) $\text{Sa}^2(\underline{4\pi t}) \text{Sa}^4(\underline{3\pi t})$

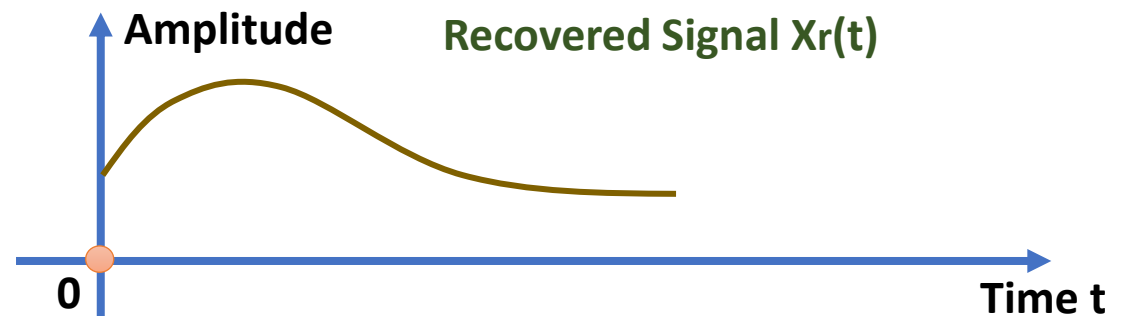
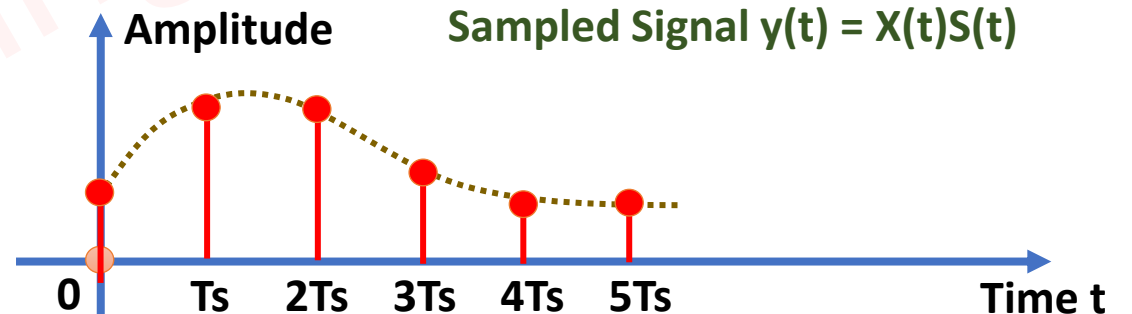
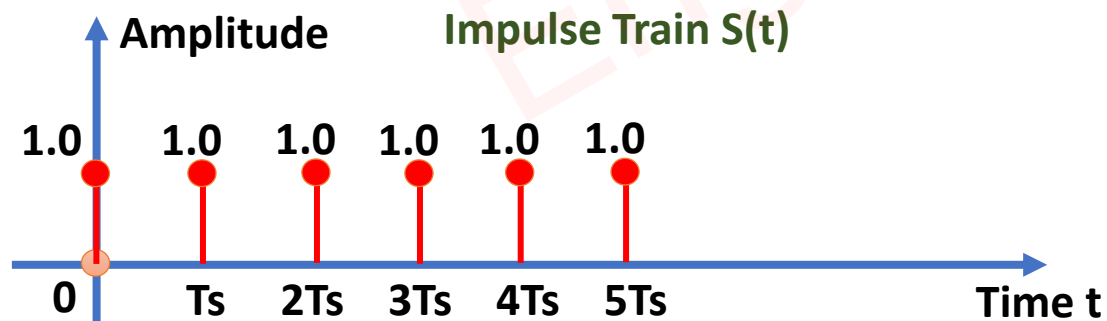
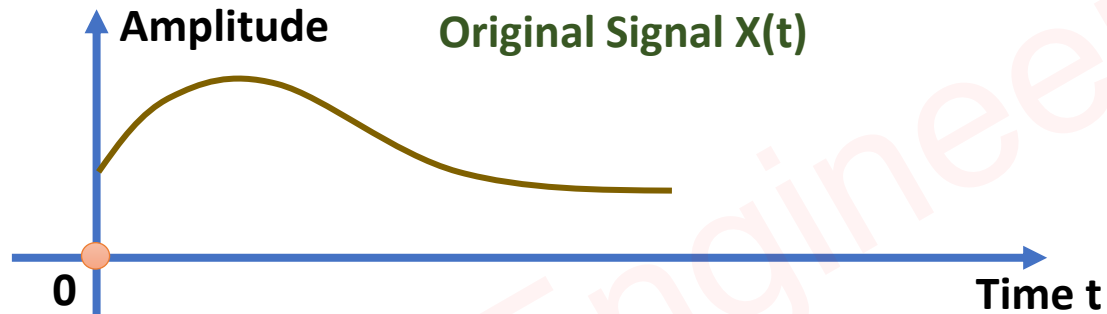
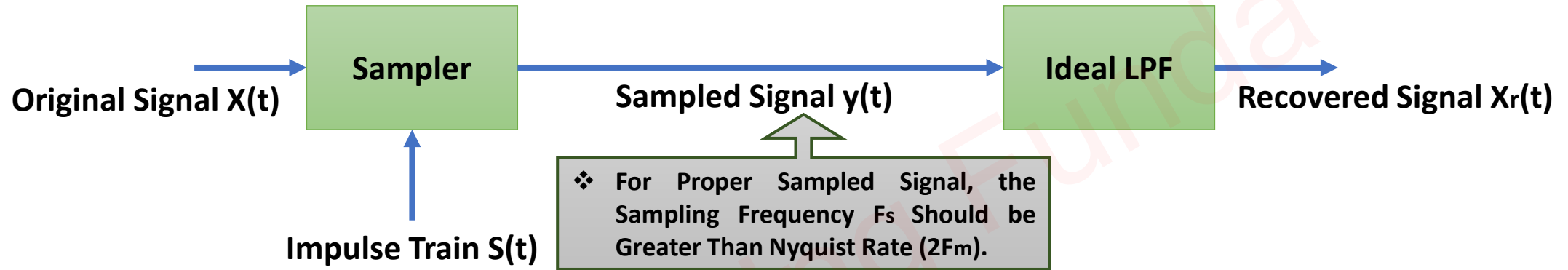
$\downarrow$
 $\omega_{m1} = 4\pi$

$\downarrow$
 $\omega_{m2} = 3\pi$

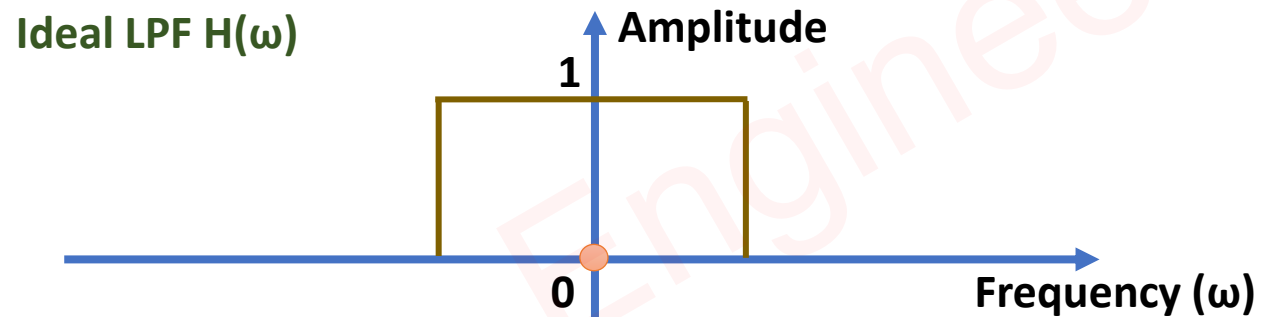
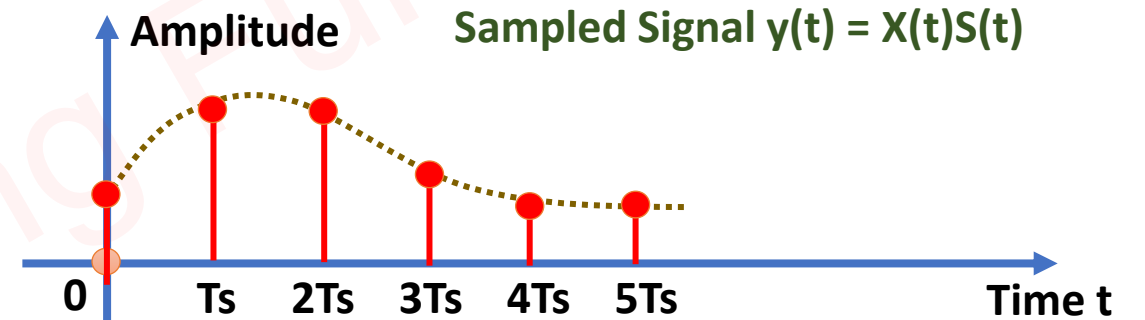
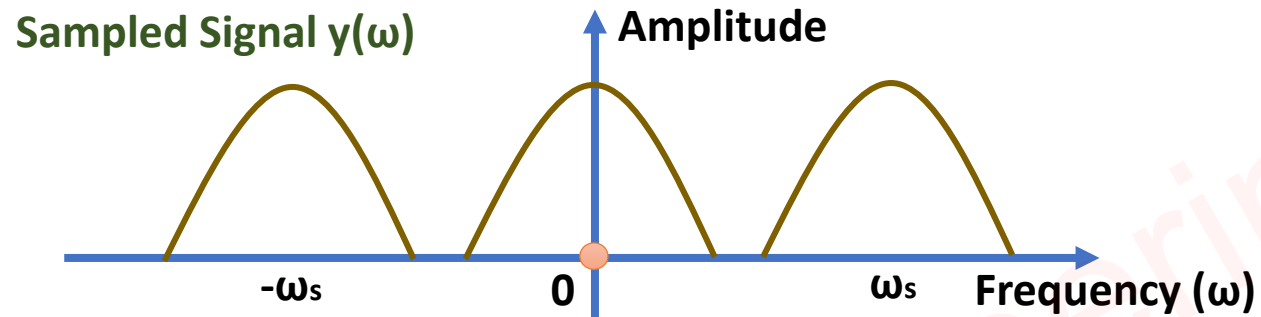
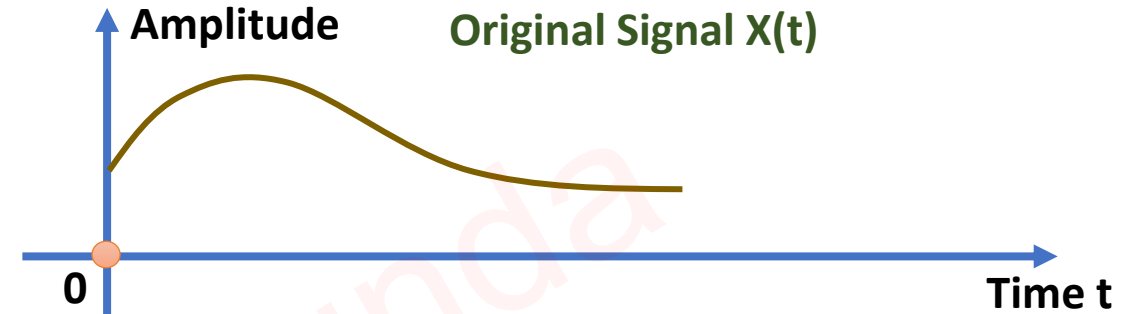
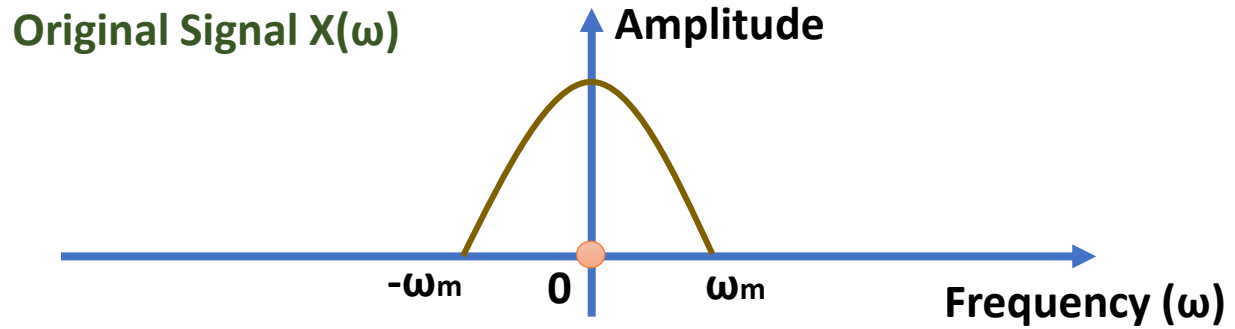
$\rightarrow \omega_{Niii} = 2(2(4\pi)) + 4(2(3\pi))$
 $= 40\pi \text{ rad/sec}$

Signal Recovery After Sampling

Signal Recovery using Ideal LPF

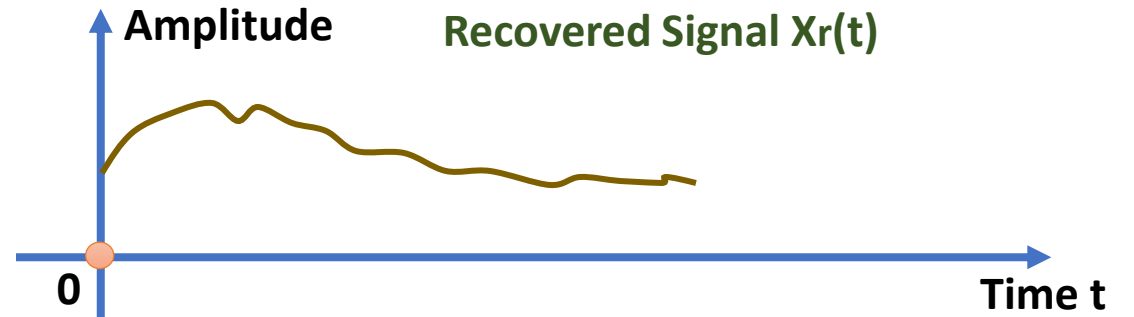
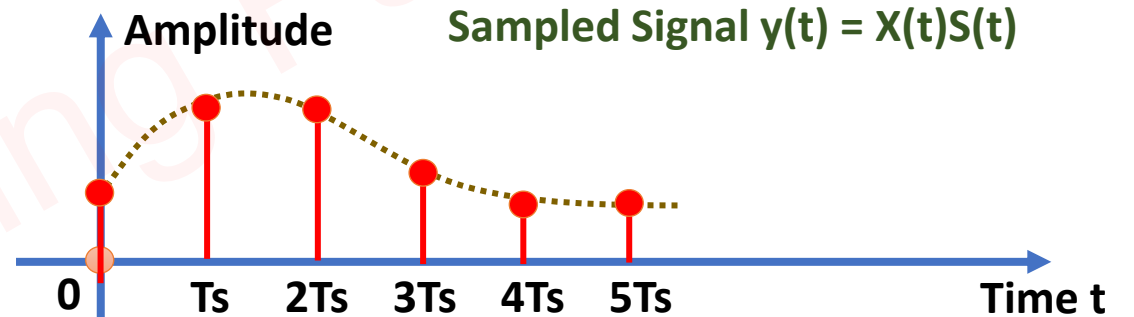
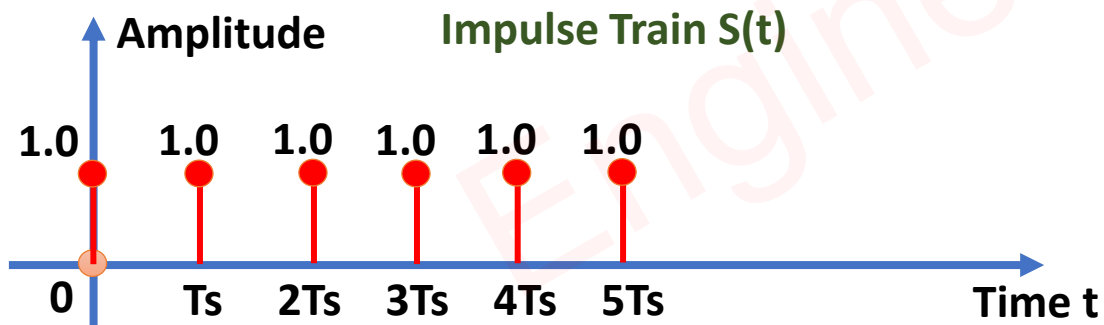
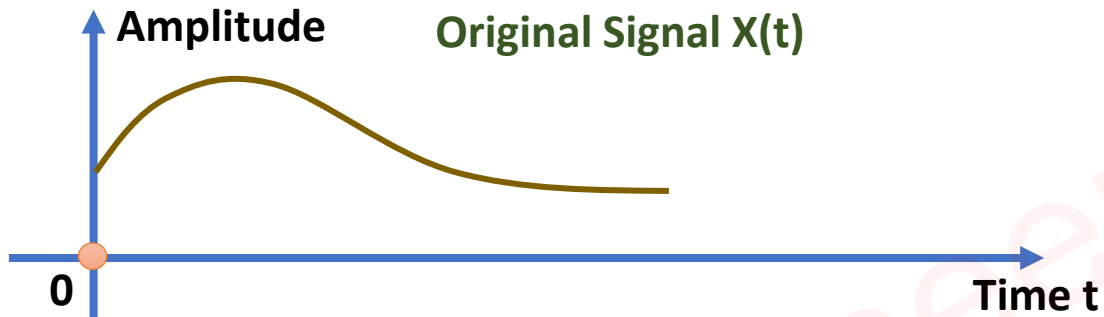
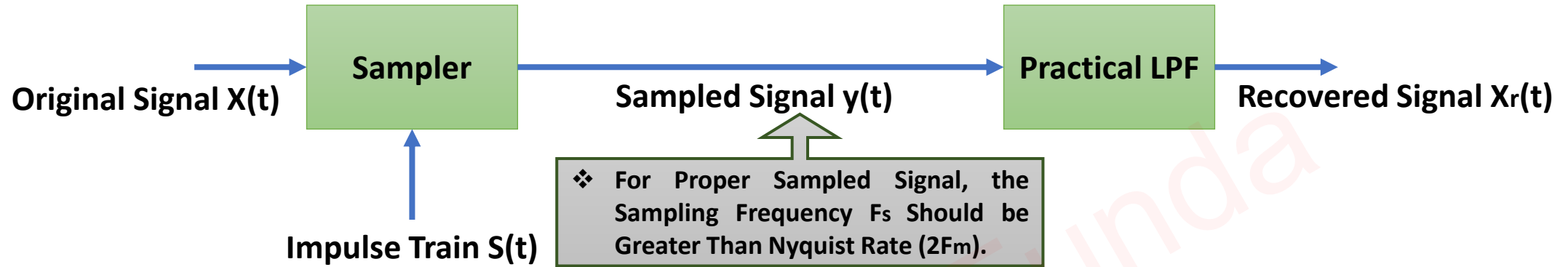


Frequency Spectrum of Sampling with Ideal LPF



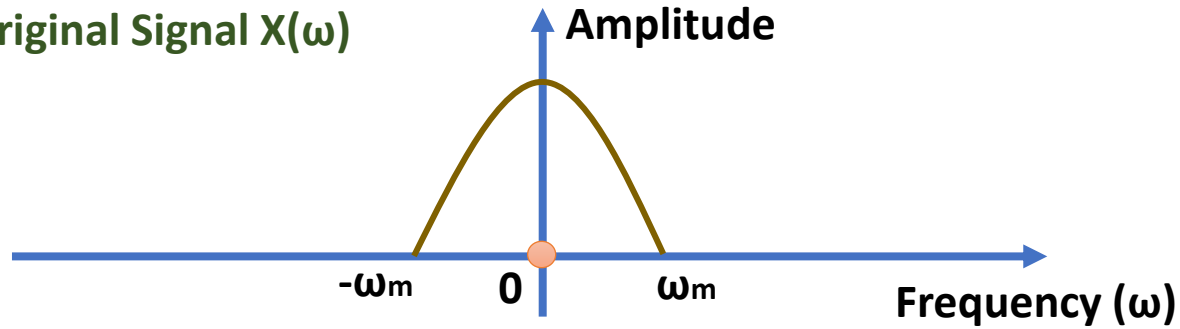
Recovered Signal $X_r(\omega) = H(\omega)y(\omega)$

Signal Recovery using Practical LPF

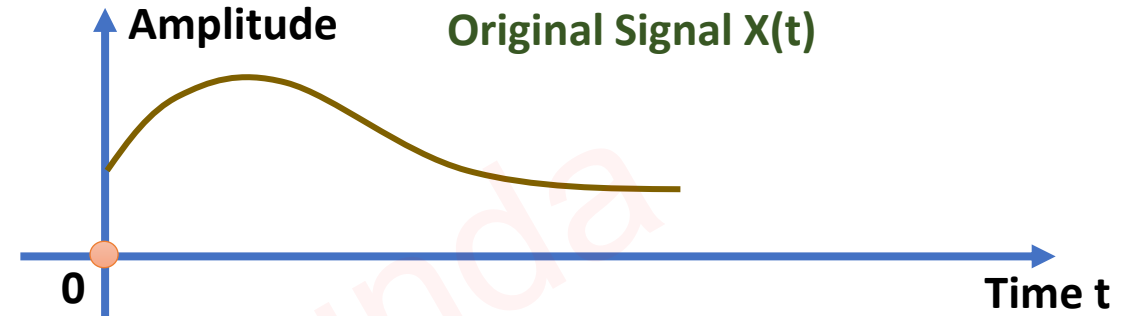


Frequency Spectrum of Sampling with Practical LPF

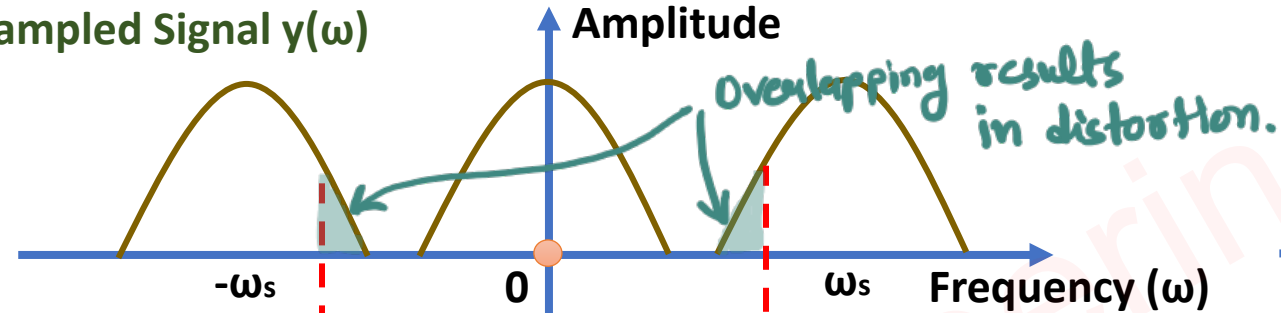
Original Signal $X(\omega)$



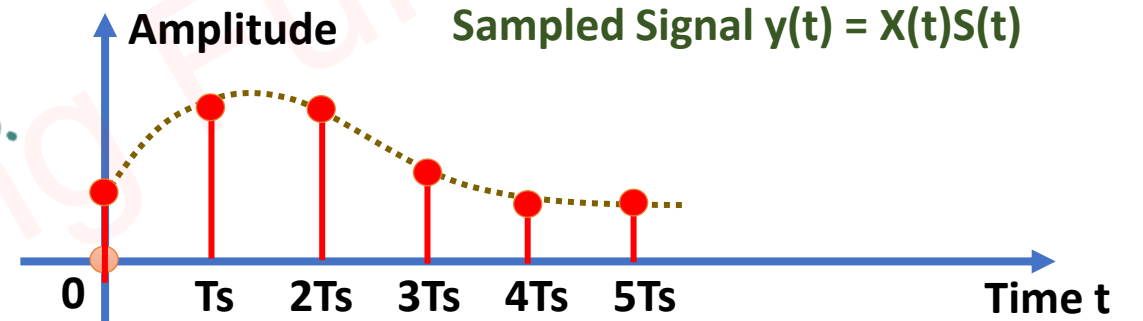
Original Signal $X(t)$



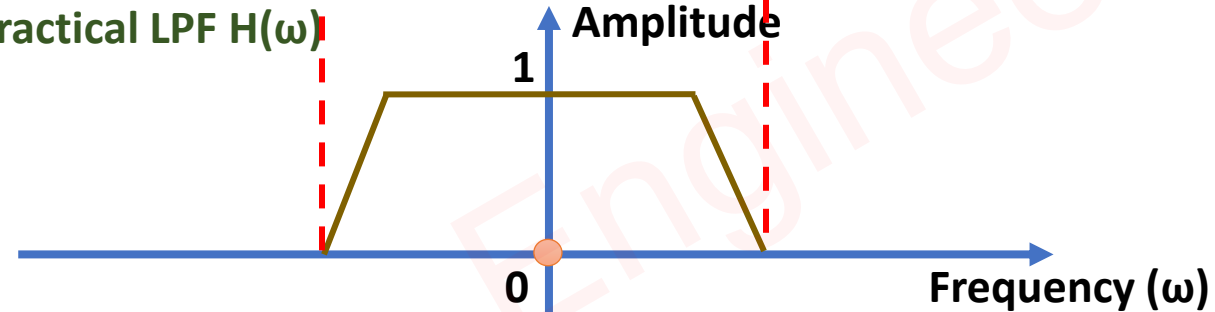
Sampled Signal $y(\omega)$



Sampled Signal $y(t) = X(t)S(t)$



Practical LPF $H(\omega)$



Recovered Signal $X_r(\omega) = H(\omega)y(\omega)$

Examples of Signal Recovery after Sampling Process

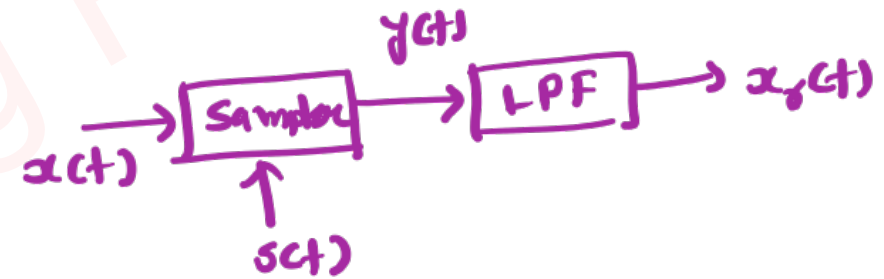
Example 1: Let $x(t) = 10 \sin(400\pi t)$ be sampled at 300Hz and reconstructed using an ideal Low Pass Filter with cut-off frequency of 150 Hz. The Frequencies present in the reconstructed signal is/are

- A. 50Hz, 100 Hz
- ✓ B. 100 Hz
- C. 100 Hz, 150 Hz
- D. 50 Hz, 100 Hz, 150 Hz

$$\rightarrow f_m = \frac{\omega_m}{2\pi} = 200 \text{ Hz}$$

$$f_s = 300 \text{ Hz}$$

$$f_c = 150 \text{ Hz}$$



$$\rightarrow y(\omega) \rightarrow x(\omega) + x(\omega \pm \omega_s) + x(\omega \pm 2\omega_s) + \dots$$

$\downarrow$	$\downarrow$	$\downarrow$
$\pm f_m$	$\pm f_s \pm f_m$	$\pm 2f_s \pm f_m$
- 200 Hz	- 500 Hz 100 Hz	

Example 2: Let $x(t) = \cos(10\pi t) + \cos(30\pi t)$ be sampled at 20Hz and reconstructed using an ideal Low Pass Filter with cut-off frequency of 20 Hz. The Frequencies present in the reconstructed signal is/are

- ✓ A. 5 Hz and 15 Hz Only
- B. 10 Hz and 15 Hz Only
- C. 5 Hz, 10, and 15 Hz Only
- D. 5 Hz Only

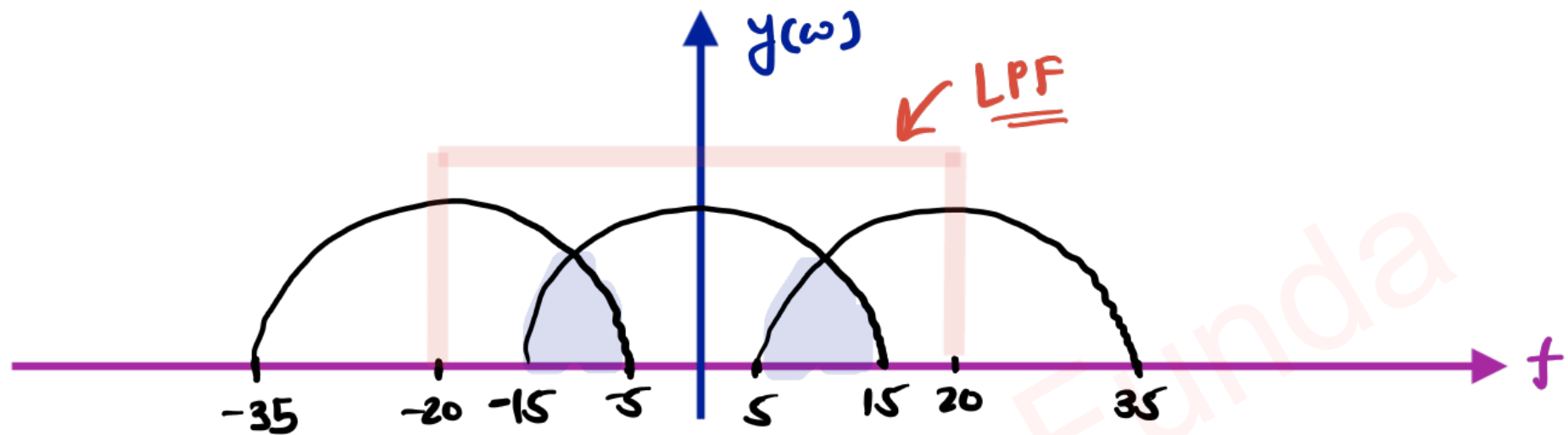
$$\sim \left. \begin{aligned} f_1 &= \frac{\omega_1}{2\pi} = 5 \text{ Hz} \\ f_2 &= \frac{\omega_2}{2\pi} = 15 \text{ Hz} \end{aligned} \right\} \underline{\underline{f_m}}$$

$$f_s = 20 \text{ Hz}$$

$$f_c = 20 \text{ Hz}$$

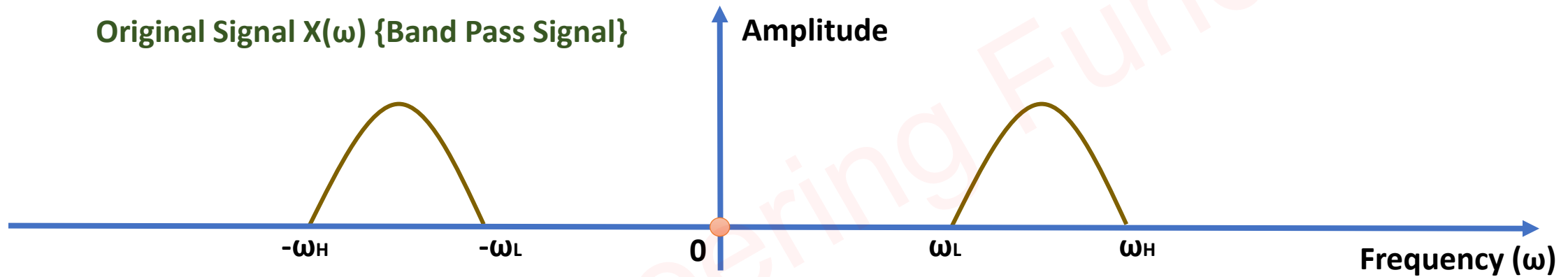
$$\sim y(\omega) \rightarrow x(\omega) + x(\omega \pm \omega_s) + x(\omega \pm 2\omega_s) + \dots$$

↓	↓	↓
$\pm f_m$	$\pm f_s \pm f_m$	$\pm 2f_s \pm f_m$
- 5 Hz	- 25 Hz	
15 Hz	- 15 Hz	
	- 35 Hz	
	- 5 Hz	



Sampling Theorem for Band Pass Signals

Characteristics of Band Pass Signals



- ❑ Band Pass Signal $X(\omega)$ has a Bandwidth $B = \omega_H - \omega_L$
- ❑ A Band Pass Signal $X(\omega)$ can be recovered from its sampled signal only if the sampling frequency is at least twice the Bandwidth of the Original Signal $X(\omega)$.

$$\Rightarrow F_s \geq 2B$$

- ❑ In Practice, Minimum Sampling Frequency $F_s = 2F_m/k$, where $k = F_m/B$.
- ❑ **Note: k should be a Minimum Integer Number**

Example 1: A Band Pass Message extends from 4kHz to 6kHz. What is the smallest sampling frequency required to retain the signal completely?

$$\leadsto F_L = 4 \text{ kHz}$$

$$F_H = 6 \text{ kHz}$$

$$\leadsto B = F_H - F_L = 2 \text{ kHz}$$

$$\leadsto k = \frac{F_m}{B} = \frac{6}{2} = 3$$

$$\begin{aligned}\leadsto F_s &= \frac{2F_m}{k} \\ &= \frac{2(6)}{3} \\ &= 4 \text{ kHz}\end{aligned}$$

Example 2: The Message Signal is given by:

$$\Rightarrow x(t) = 10[\cos(2000\pi t) \cos(8000\pi t)]$$

1. What is the minimum sampling rate based on the Low Pass uniform Sampling theorem?
2. What is the minimum sampling rate based on the Band Pass Sampling theorem?

$$\begin{aligned} \leadsto x(t) &= 10 [\cos(2000\pi t) \cos(8000\pi t)] \\ &= 5 [\cos(10000\pi t) + \cos(6000\pi t)] \end{aligned}$$

$$\begin{aligned} \leadsto F_1 &= \frac{\omega_1}{2\pi} = 5 \text{ KHz} \\ F_2 &= \frac{\omega_2}{2\pi} = 3 \text{ KHz} \end{aligned} \quad \left. \vphantom{\begin{aligned} F_1 &= 5 \text{ KHz} \\ F_2 &= 3 \text{ KHz} \end{aligned}} \right\} \begin{aligned} F_m &= 5 \text{ KHz} \\ B &= F_1 - F_2 \\ &= 2 \text{ KHz} \end{aligned}$$

$$\leadsto F_s = 2F_m \\ = 10 \text{ KHz}$$

← LPF

$$\begin{aligned} \leadsto K &= F_m/B \\ &= 5/2 \\ &= 2.5 \\ &\approx 2 \end{aligned}$$

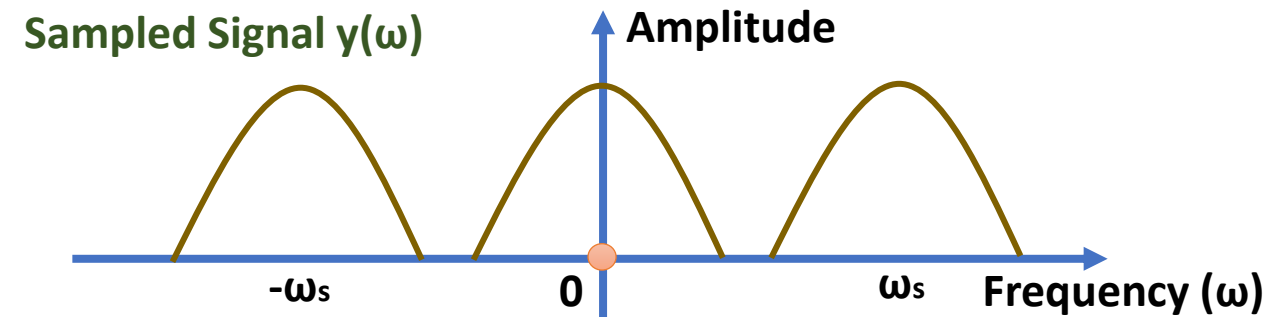
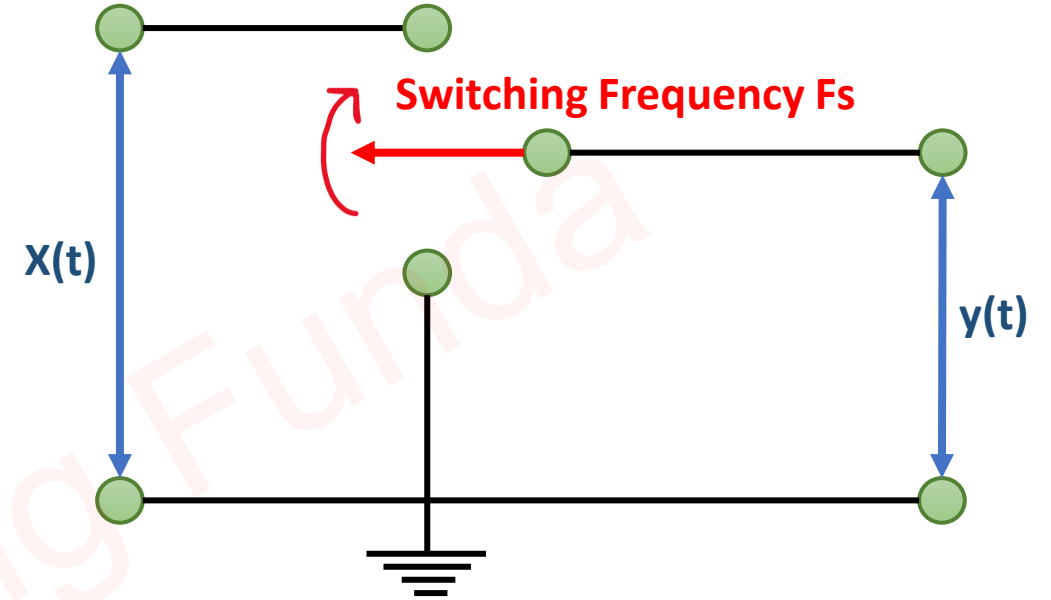
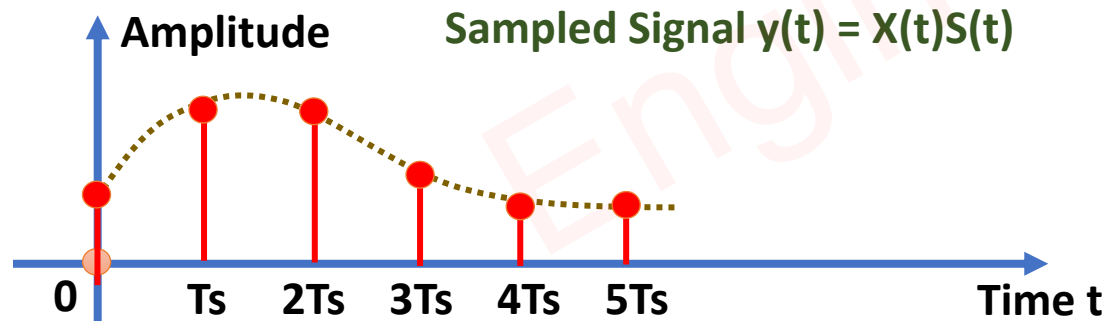
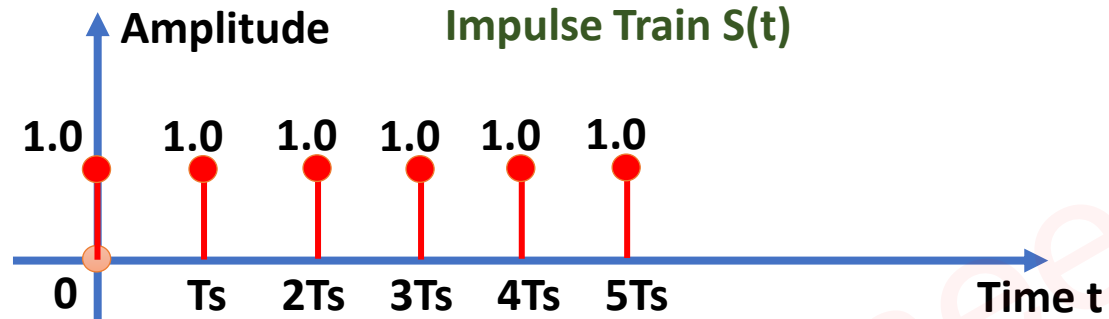
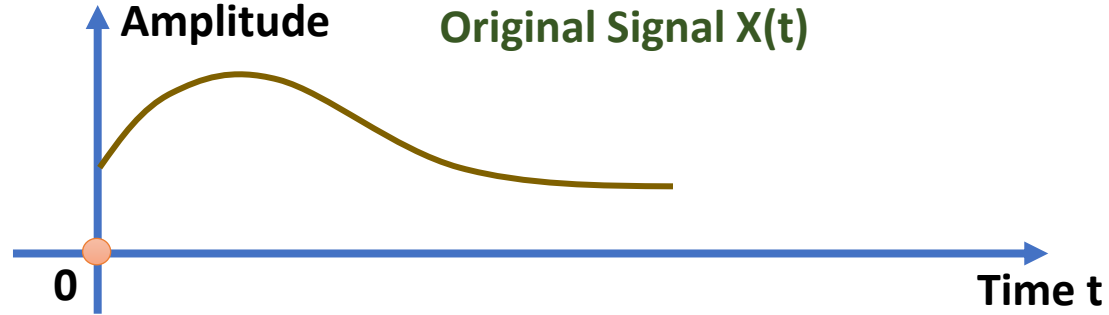
$$\begin{aligned} \leadsto F_{s(BPS)} &= 2F_m/K \\ &= 10/2 \\ &= 5 \text{ KHz} \\ &\quad \uparrow \\ &\quad \text{BPS} \end{aligned}$$

Ideal Sampling or Instantaneous Sampling or Impulse Sampling

Outlines

- ☐ Waveforms of Ideal Sampling
- ☐ Calculation of Ideal Sampling
- ☐ Basics of Ideal Sampling
- ☐ Key Points of Ideal Sampling

Waveforms of Ideal Sampling



Calculation of Ideal Sampling

- Impulse Train at Sampling interval T_s defined as:

$$\Rightarrow s(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_s) = \frac{1}{T_s} [1 + 2 \cos(\omega_s t) + 2 \cos(2\omega_s t) + \dots]$$

- Output of Ideal Sampling in Time domain is given by $y(t) = X(t)s(t)$

$$\Rightarrow y(t) = \frac{1}{T_s} X(t) [1 + 2 \cos(\omega_s t) + 2 \cos(2\omega_s t) + \dots]$$

- Output of Ideal Sampling in the Frequency domain is given by $y(\omega)$

$$\Rightarrow y(\omega) = \frac{1}{T_s} [X(\omega) + X(\omega \pm \omega_s) + X(\omega \pm 2\omega_s) + \dots]$$

Basics of Ideal Sampling

- ❑ To generate an ideal sample train, we use a switching sampler.
- ❑ Closing time $t \rightarrow 0$, then only an ideal impulse train is possible.
- ❑ Sampling Frequency F_s should be Greater than or equal to the Nyquist Rate $2F_m$.

Key Points of Ideal Sampling

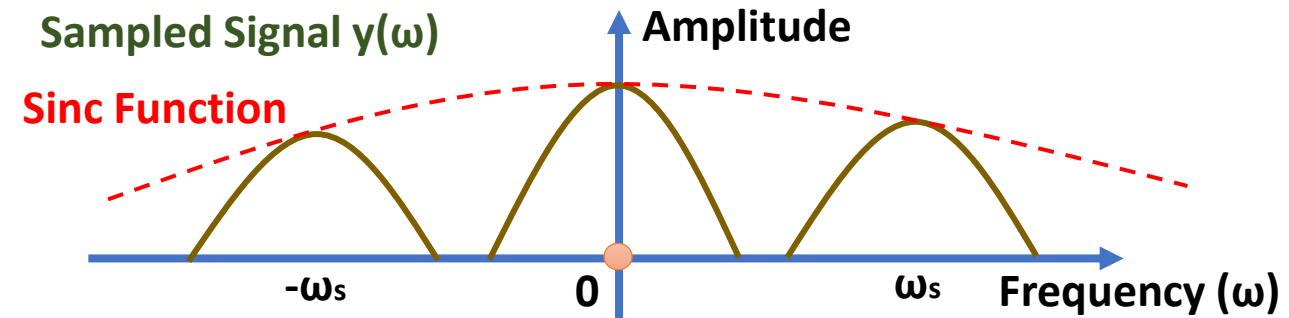
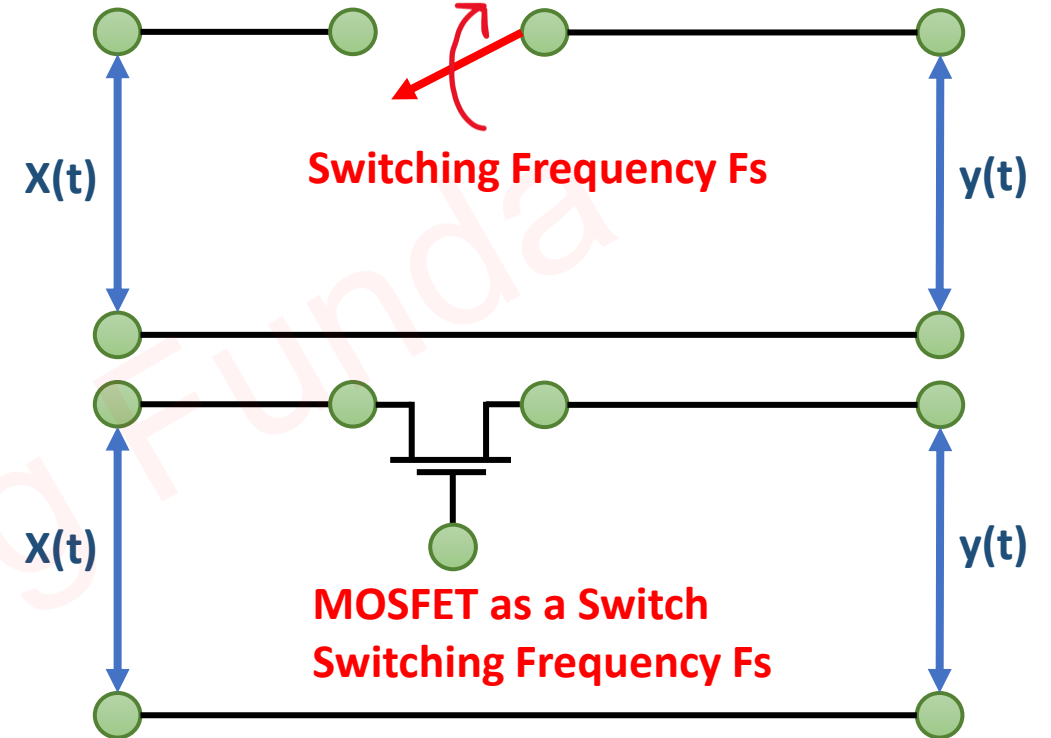
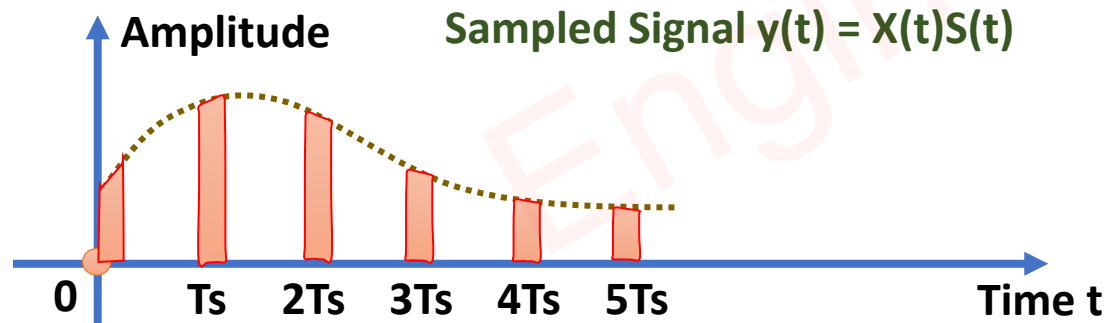
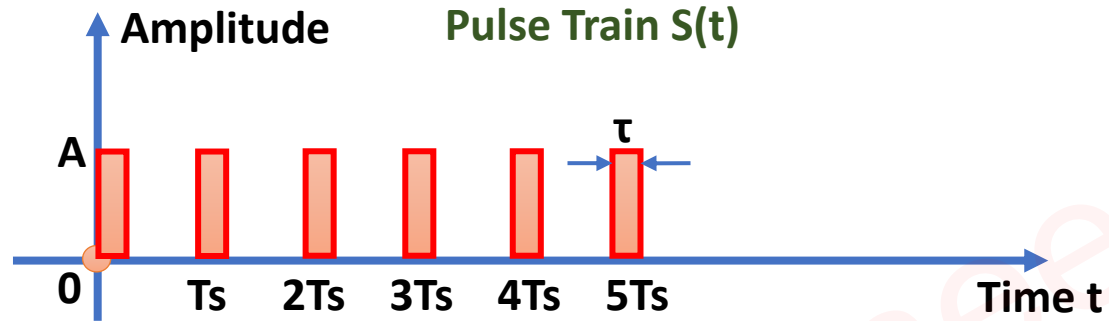
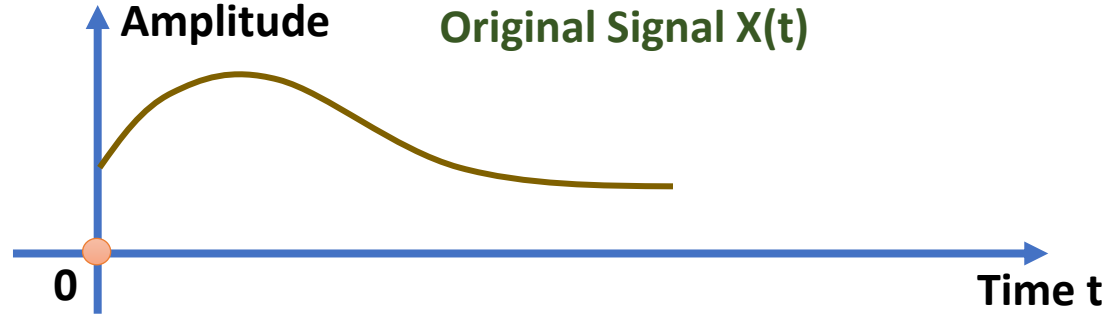
- ❑ Practically, it is not possible.
- ❑ It has high Noise Interference.

Natural Sampling or Chopper Sampling or Ordinary Sampling

Outlines

- ☐ Waveforms of Natural Sampling
- ☐ Calculation of Natural Sampling
- ☐ Basics of Natural Sampling

Waveforms of Natural Sampling



Calculation of Natural Sampling

□ Pulse Train at Sampling interval T_s and Pulse width τ defined as:

$$\Rightarrow s(t) = \frac{\tau A}{T_s} \sum_{n=-\infty}^{\infty} \text{sinc}(f_n \tau) e^{j2\pi n f_s t}$$

□ Output of Natural Sampling in Time domain is given by $y(t) = X(t)s(t)$

$$\Rightarrow y(t) = \frac{\tau A}{T_s} \sum_{n=-\infty}^{\infty} X(t) \text{sinc}(f_n \tau) e^{j2\pi n f_s t}$$

□ Output of Natural Sampling in the Frequency domain is given by $y(f)$

$$\Rightarrow y(f) = \frac{\tau A}{T_s} \sum_{n=-\infty}^{\infty} X(f - n f_s) \text{sinc}(n f_s \tau)$$

Basics of Natural Sampling

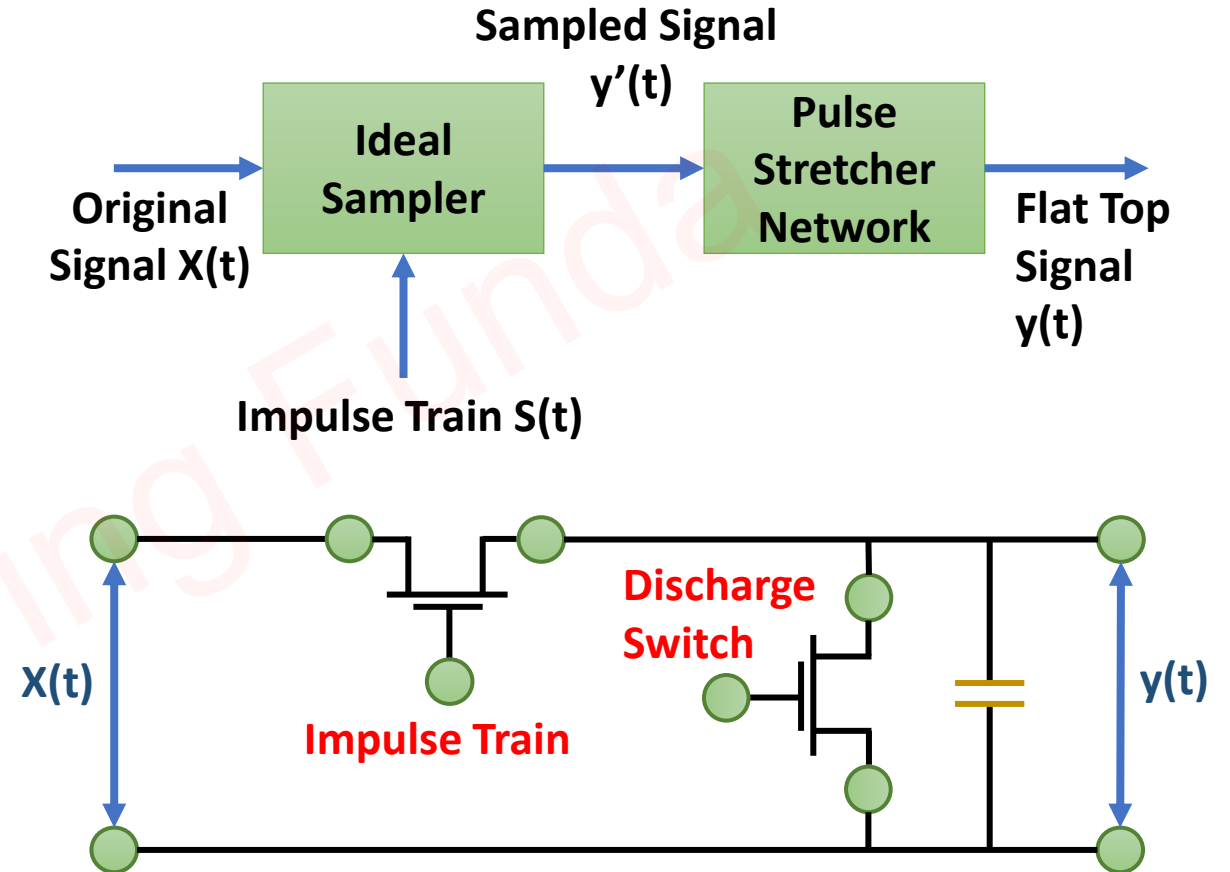
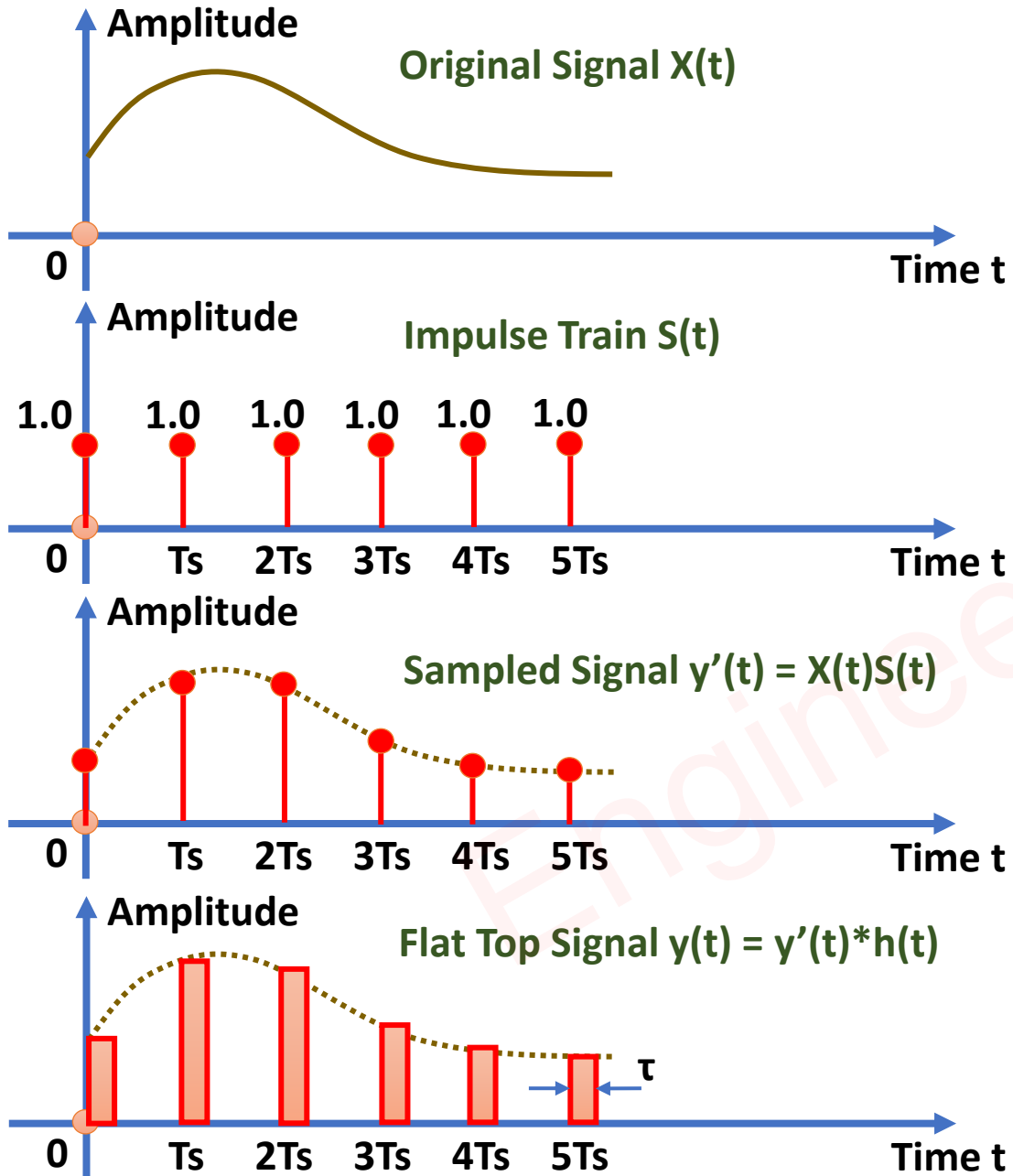
- ❑ To generate a Natural sampling, we use a switching sampler/ MOSFET Switch.
- ❑ Closing time $t = \tau$.
- ❑ Sampling Frequency F_s should be Greater than or equal to the Nyquist Rate $2F_m$.
- ❑ Here, we can use a practical LPF to reconstruct the Sampled Signal.
- ❑ For a Large value of τ , there is a possibility of crosstalk.

Flat Top Sampling/ Sample & Hold Sampling/ Rectangular Pulse Sampling/ PAM

Outlines

- ☐ Waveforms of Flat Top Sampling
- ☐ Calculation of Flat Top Sampling
- ☐ Aperture Effect in Flat Top Sampling
- ☐ Basics of Flat Top Sampling

Waveforms of Flat Top Sampling



Calculation of Flat Top Sampling

- Impulse Train at Sampling interval T_s defined as:

$$\Rightarrow S(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_s) = \frac{1}{T_s} [1 + 2 \cos(\omega_s t) + 2 \cos(2\omega_s t) + \dots]$$

- Output of Ideal Sampling in the Time domain is given by $y'(t) = X(t)s(t)$

$$\Rightarrow y'(t) = \sum_{n=-\infty}^{\infty} X(t)\delta(t - nT_s)$$

- Output of Ideal Sampling in the Frequency domain is given by $y'(f)$

$$\Rightarrow y'(f) = \frac{1}{T_s} \sum_{n=-\infty}^{\infty} X(f - nf_s)$$

- Output of Flat Top Sampling in the Time domain is given by $y(t) = y'(t)*h(t)$

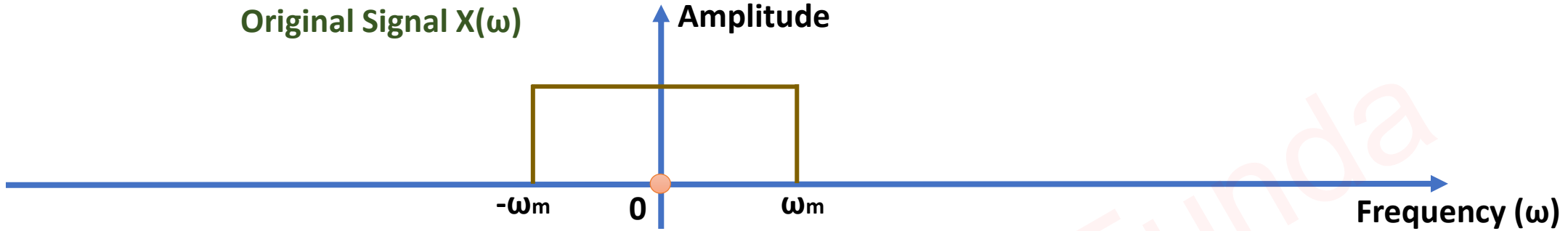
$$\Rightarrow y(t) = \sum_{n=-\infty}^{\infty} X(nT_s)h(t - nT_s)$$

- Output of Flat Top Sampling in the Frequency domain is given by $y(f)$

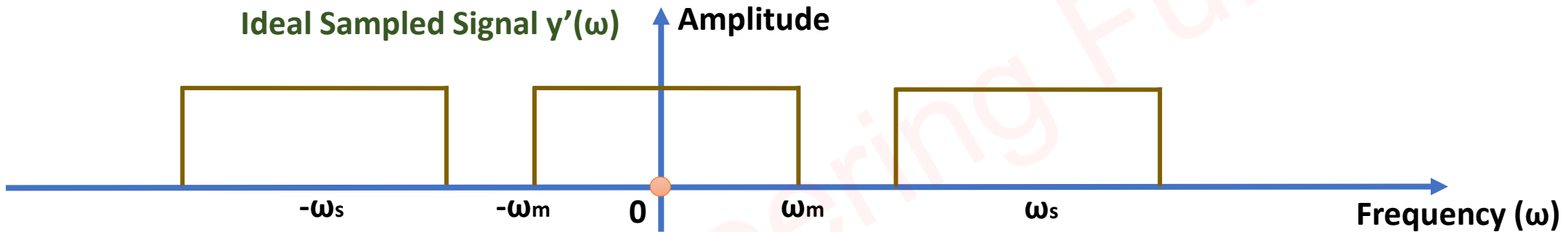
$$\Rightarrow y(f) = \frac{1}{T_s} \sum_{n=-\infty}^{\infty} X(f - nf_s)H(f)$$

Aperture Effect in Flat Top Sampling

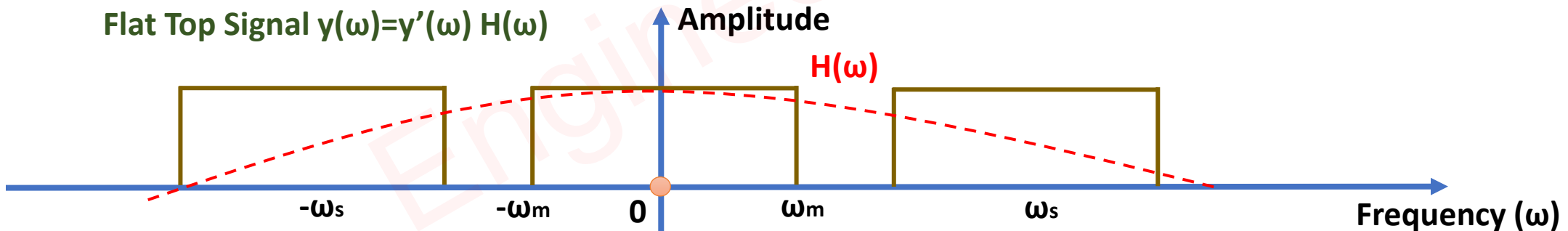
Original Signal $X(\omega)$



Ideal Sampled Signal $y'(\omega)$



Flat Top Signal $y(\omega) = y'(\omega) H(\omega)$



Basics of Flat Top Sampling

- ❑ Pulse Stretcher Network (Pulse Network) is a Sinc Function in the frequency domain.
- ❑ Due to the Pulse Stretcher Network and Ideal Sampling combination, there is a loss of high-frequency components in the message signal, which is known as the Aperture Effect.
- ❑ If we increase the Pulse width τ , there will be an increase in the Aperture effect.

Ideal Vs Natural Vs Flat Top Sampling

Parameters	Ideal Sampling	Natural Sampling	Flat Top Sampling
Sampling Principle	❖ Train of Impulses (Multiplication)	❖ Train of Finite Duration of Pulses (Chopping Principle)	❖ Train of Finite Duration of Pulses (Sample & Hold)
Circuit			
Waveforms			
Practical Realization	❖ Not Possible	❖ Practical method	❖ Practical Method

Parameters	Ideal Sampling	Natural Sampling	Flat Top Sampling
Sampling Rate	❖ Can be Infinity	❖ Should Satisfy Nyquist Rate	❖ Should Satisfy Nyquist Rate
Noise Interference	❖ Very High	❖ Moderate	❖ Moderate to High
Time Domain Output	$\Rightarrow y(t) = \sum_{n=-\infty}^{\infty} X(t)\delta(t - nT_s)$	$\Rightarrow y(t) = \frac{\tau A}{T_s} \sum_{n=-\infty}^{\infty} X(t) \frac{\text{sinc}(nf_s \tau)}{e^{j2\pi n f_s t}}$	$\Rightarrow y(t) = \sum_{n=-\infty}^{\infty} X(nT_s)h(t - nT_s)$
Frequency Domain Output	$\Rightarrow y(f) = \frac{1}{T_s} \sum_{n=-\infty}^{\infty} X(f - nf_s)$	$\Rightarrow y(f) = \frac{\tau A}{T_s} \sum_{n=-\infty}^{\infty} X(f - nf_s) \text{sinc}(nf_s \tau)$	$\Rightarrow y(f) = \frac{1}{T_s} \sum_{n=-\infty}^{\infty} X(f - nf_s)H(f)$
Bandwidth Requirement	❖ Very High	❖ Increase with reduction in pulse width τ .	❖ Increase with reduction in pulse width τ .

Examples of Sampling & Nyquist Rate

Example 1: An Analog Signal is given as:

$$\Rightarrow x(t) = \sin(150\pi t) + 2 \sin(300\pi t) + 2 \cos(600\pi t)$$

Find Nyquist Rate and Nyquist Interval for given Signal. Does aliasing occurs, if signal is sampled at 400 Hz.

$$\sim f_1 = \frac{\omega_1}{2\pi} = 75 \text{ Hz}$$

$$f_2 = \frac{\omega_2}{2\pi} = 150 \text{ Hz}$$

$$f_3 = \frac{\omega_3}{2\pi} = 300 \text{ Hz}$$

→ Max. freq.

$$f_m = 300 \text{ Hz}$$

$$\sim F_N = 2f_m \\ = 600 \text{ Hz}$$

$$\sim T_N = 1/F_N = 1.66 \text{ msec}$$

→ Aliasing check.

- $F_1 = 75 \text{ Hz} \rightarrow NR = 150 \text{ Hz} < F_s \rightarrow \text{No Aliasing}$

$F_2 = 150 \text{ Hz} \rightarrow NR = 300 \text{ Hz} < F_s \rightarrow \text{No Aliasing}$

$F_3 = 300 \text{ Hz} \rightarrow NR = 600 \text{ Hz} > F_s \rightarrow \checkmark \text{ Aliasing}$

Example 2: An Analog Signal is given as:

$$\Rightarrow x(t) = \cos(200\pi t) + 0.25 \cos(700\pi t)$$

if signal is sampled at 400 samples/sec. Sampled waveform is passed through ideal LPF with cut-off frequency 200Hz. Sketch the spectrum.

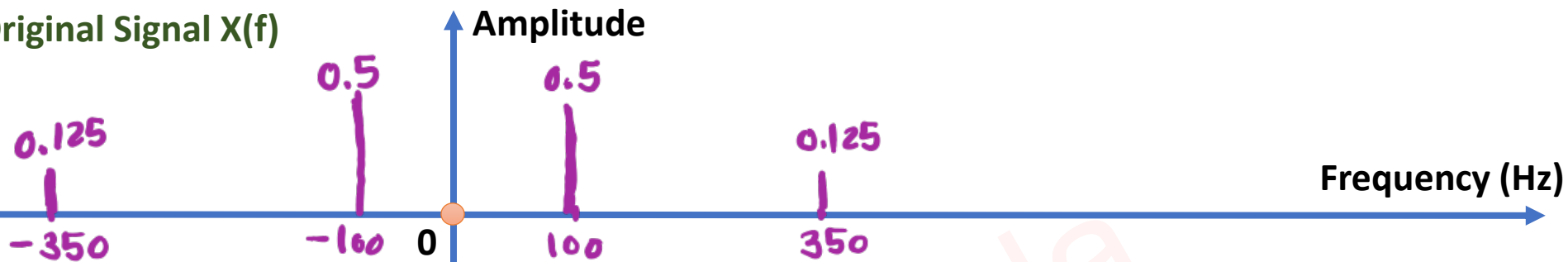
$$\leadsto A \cos(2\pi f_0 t) \xrightarrow{\text{F.T.}} \frac{A}{2} [\delta(f - f_0) + \delta(f + f_0)]$$

$$\leadsto x(f) = \frac{1}{2} [\delta(f - 100) + \delta(f + 100)] + 0.125 [\delta(f - 350) + \delta(f + 350)]$$

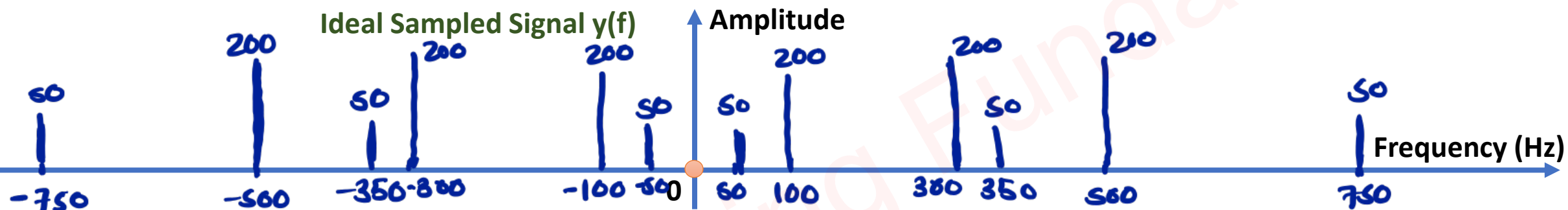
$$\begin{aligned} \leadsto y(f) &= \frac{1}{T_s} \sum_{n=-\infty}^{\infty} x(f - nF_s) \\ &= F_s [\dots + x(f - F_s) + x(f) + x(f + F_s) + \dots] \end{aligned}$$

$\nearrow$
400 Hz

Original Signal $X(f)$



Ideal Sampled Signal $y(f)$



Low Pass Filter $h(f)$



O/P of Low Pass Filter $y'(f)$

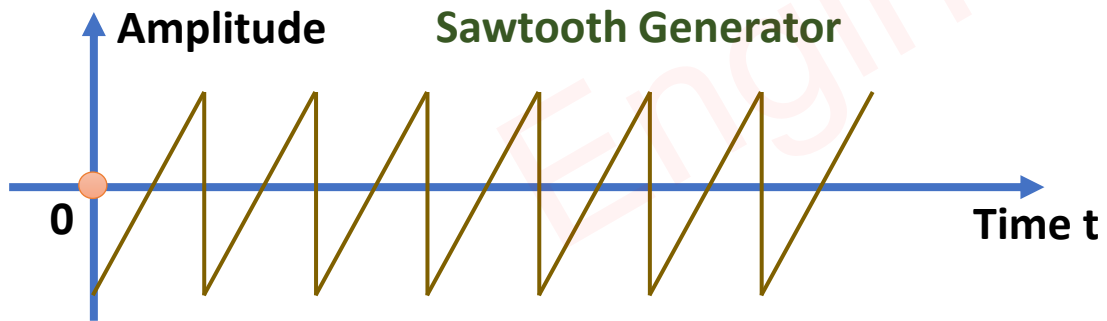
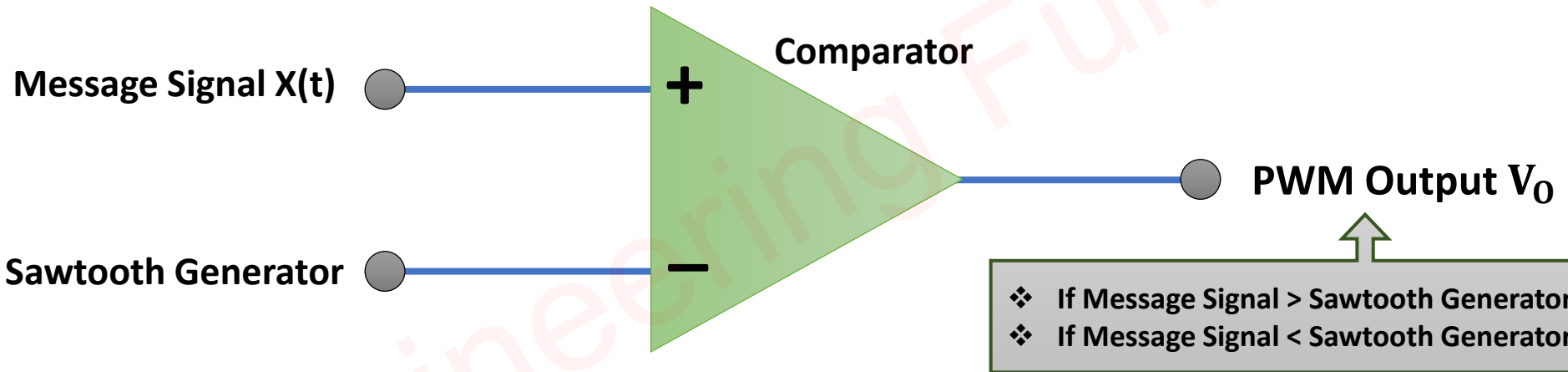
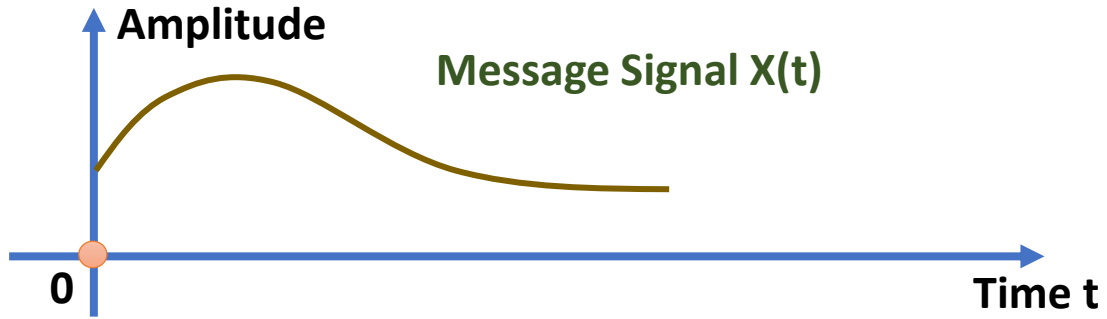


Pulse Width Modulation PWM

Outlines

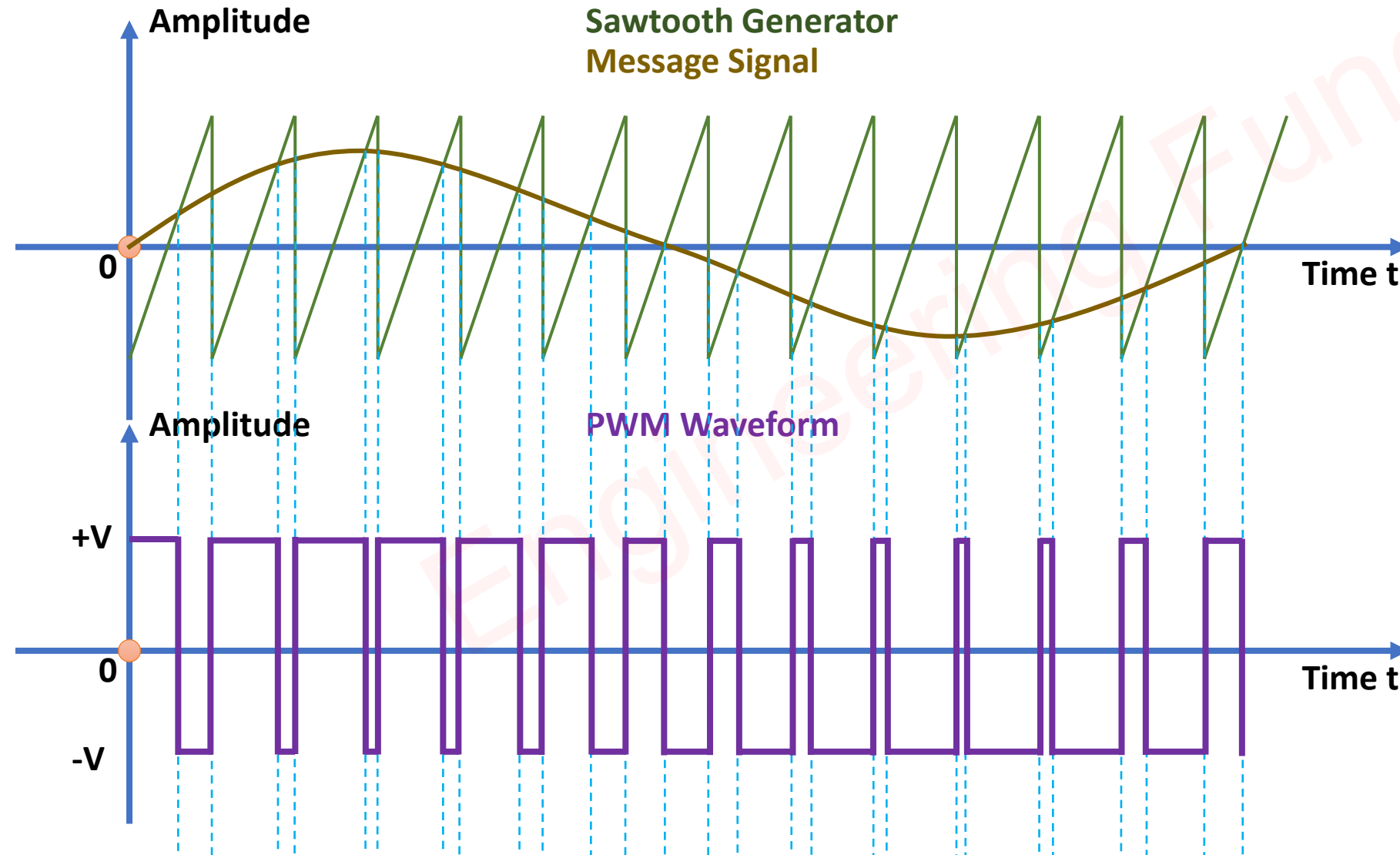
- ❑ PWM Circuit
- ❑ Waveforms of PWM
- ❑ PWM Circuit with Negative Clipper
- ❑ Basics of PWM
- ❑ Parameters of PWM
- ❑ Applications of PWM

PWM Circuit

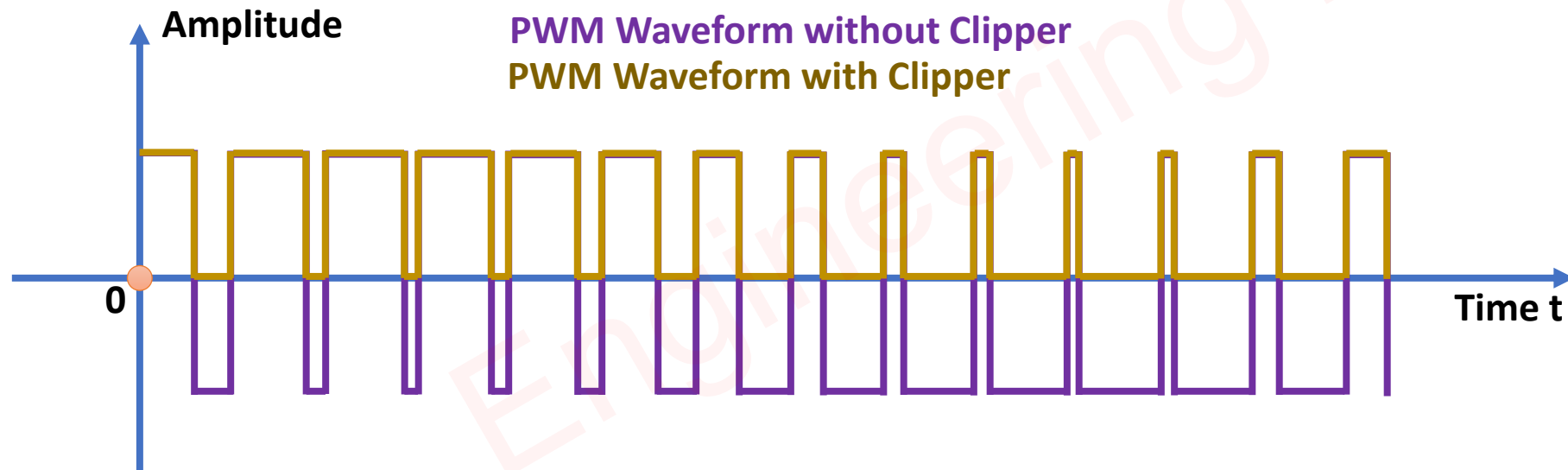
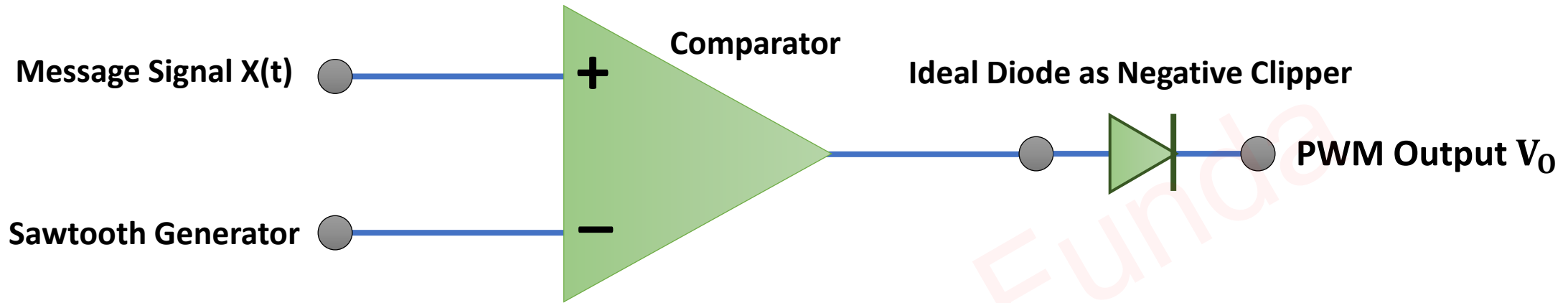


PWM Waveforms

- ❖ If Message Signal > Sawtooth Generator, O/P = +V
- ❖ If Message Signal < Sawtooth Generator, O/P = -V



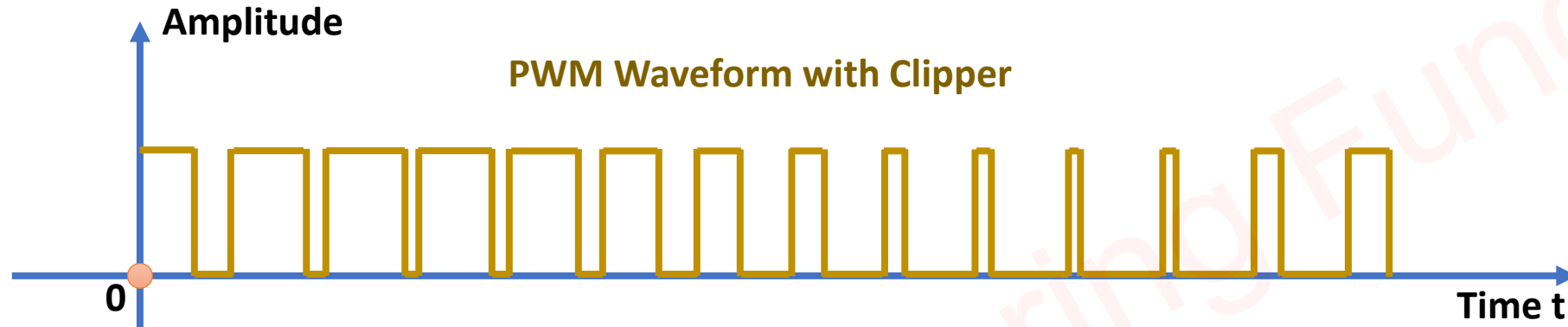
PWM Circuit with Negative Clipper



❑ PWM with a negative clipper is widely used for Motor Drives.

Basics of PWM

- ❑ PWM is a technique where the width of the pulse (ON time) is varied while keeping the frequency constant.
- ❑ PWM main Parameters are Duty Cycle D and Frequency F.



$$\Rightarrow D = \frac{T_{ON}}{T_{ON} + T_{OFF}} = \frac{T_{ON}}{T}$$

$$\Rightarrow F = \frac{1}{T}$$

- ❑ In PWM, the duty cycle (D) varies according to the message signal while the frequency (F) is kept constant.

Applications of PWM

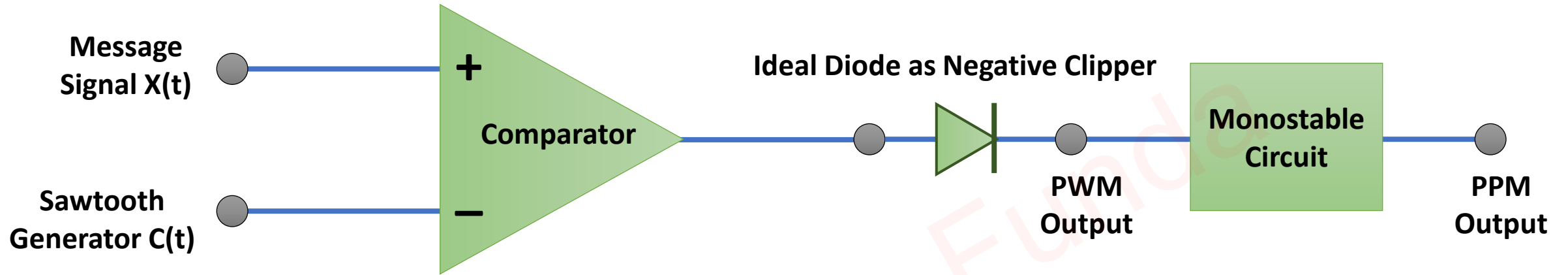
- ❑ **Motor Control:** {Speed control of DC motors and Torque control in AC drives.}
- ❑ **Power Regulation:** {Voltage regulators and Dimming of LEDs.}
- ❑ **Signal Processing:** {Audio amplifiers (Class-D) and Digital-to-Analog conversion.}
- ❑ **Communication:** {Pulse modulation schemes.}
- ❑ **Embedded Systems:** {Microcontroller-based servo control and Power management in devices.}

Pulse Position Modulation PPM

Outlines

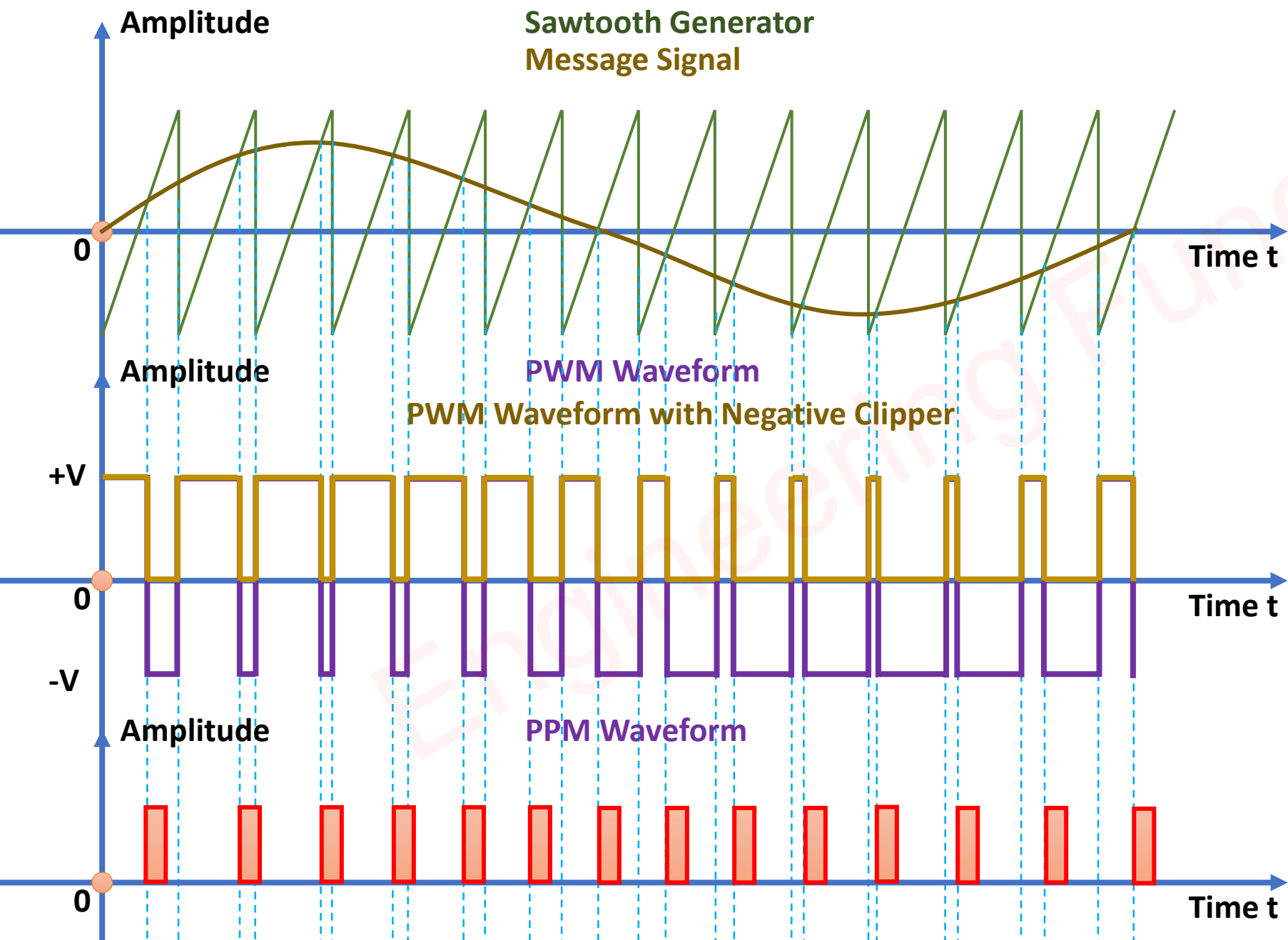
- ❑ PPM Circuit
- ❑ Waveforms of PPM
- ❑ Basics of PPM
- ❑ Advantages of PPM
- ❑ Disadvantages of PPM
- ❑ Applications of PPM

PPM Circuit



- ❑ In PPM generation, a PWM signal is first produced, and after passing it through a monostable circuit, a PPM signal is obtained.
- ❑ Here, the monostable circuit generates a fixed-duration pulse at the falling edge of the PWM signal.

PPM Waveforms



Basics of PPM

- ❑ Pulse Position Modulation (PPM) is a modulation technique where the position of a pulse (within a fixed time frame) is varied according to the amplitude of the modulating signal.
- ❑ Characteristics of Pulse in PPM:
 - Fixed Duration
 - Fixed Amplitude
 - Change in position with respect to the message Signal.

Advantages of PPM

- ❑ High Noise Immunity
- ❑ Efficient Power (Constant Power with respect to Time)
- ❑ Better performance over long distance than PAM and PWM.

Disadvantages of PPM

- ❑ Complex Tx and Rx Circuit.
- ❑ Timing synchronisation required.

Applications of PPM

- ❑ Remote Control Systems {e.g. RC Aircraft, Drones}
- ❑ Optical Communication

PAM Vs PWM Vs PPM

Parameters	PAM	PWM	PPM
Modulation Principle	❖ Amplitude of Pulse changes with respect to Message Signal.	❖ Width of Pulse changes with respect to Message Signal.	❖ Position of Pulse changes with respect to Message Signal.
Pulse Amplitude	❖ Changes	❖ Constant	❖ Constant
Pulse Width	❖ Constant	❖ Changes	❖ Constant
Pulse Position	❖ Constant	❖ Changes	❖ Changes
Bandwidth	❖ High	❖ Moderate	❖ Low
Synchronization	❖ Not Critical	❖ Not Critical	❖ Critical
Power Efficiency	❖ Low	❖ Moderate	❖ High
Noise Immunity	❖ Low	❖ Moderate	❖ High
Tx/Rx Complexity	❖ Low	❖ Moderate	❖ High
Applications	❖ Ethernet, DSL	❖ Motor Control	❖ Remote Control